



College of Liberal Arts & Sciences
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I wish to be considered for the PECASE honorary award.

Sincerely,

A handwritten signature in black ink, appearing to read "Peter Kvam", with a long horizontal stroke extending to the right.

Peter Kvam, Ph.D.
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COVER SHEET FOR PROPOSAL TO THE NATIONAL SCIENCE FOUNDATION

PROGRAM ANNOUNCEMENT/SOLICITATION NO./DUE DATE NSF 22-586 07/27/2022		☐ Special Exception to Deadline Date Policy		FOR NSF USE ONLY NSF PROPOSAL NUMBER 2237119	
FOR CONSIDERATION BY NSF ORGANIZATION UNIT(S) (Indicate the most specific unit known, i.e. program, division, etc.) SES - Decision, Risk & Mgmt Sci					
DATE RECEIVED	NUMBER OF COPIES	DIVISION ASSIGNED	FUND CODE	UEI (Unique Entity Identifier)	FILE LOCATION
07/25/2022	1	04050000 SES	1321	NNFQH1JAPEP3	
EMPLOYER IDENTIFICATION NUMBER (EIN) OR TAXPAYER IDENTIFICATION NUMBER (TIN) 596-00-2052		SHOW PREVIOUS AWARD NO. IF THIS IS ☐ A RENEWAL ☐ AN ACCOMPLISHMENT-BASED RENEWAL		IS THIS PROPOSAL BEING SUBMITTED TO ANOTHER FEDERAL AGENCY? YES ☐ NO ☒ IF YES, LIST ACRONYM(S)	
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TITLE OF PROPOSED PROJECT CAREER: An accessible, dynamic model of pricing applied to wage and housing purchases					SHOW LETTER OF INTENT ID IF APPLICABLE
REQUESTED AMOUNT \$ 696,802	PROPOSED DURATION (1-60 MONTHS) 60 months	REQUESTED STARTING DATE 05/01/2023	SHOW RELATED PRELIMINARY PROPOSAL NO. IF APPLICABLE		
THIS PROPOSAL INCLUDES ANY OF THE ITEMS LISTED BELOW ☐ BEGINNING INVESTIGATOR ☐ DISCLOSURE OF LOBBYING ACTIVITIES ☐ PROPRIETARY & PRIVILEGED INFORMATION ☐ HISTORIC PLACES ☐ VERTEBRATE ANIMALS IACUC App. Date _____ PHS Animal Welfare Assurance Number _____ ☒ TYPE OF PROPOSAL Research					
☒ HUMAN SUBJECTS Human Subjects Assurance Number _____ Exemption Subsection pending or IRB App. Date _____ ☐ FUNDING OF INT'L BRANCH CAMPUS OF U.S IHE ☐ FUNDING OF FOREIGN ORGANIZATION OR FOREIGN INDIVIDUAL ☐ INTERNATIONAL ACTIVITIES: COUNTRY/COUNTRIES INVOLVED _____					
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PI/PD FAX NUMBER					
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CO-PI/PD					
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By electronically signing and submitting this proposal, the Authorized Organizational Representative (AOR) is: (1) certifying that statements made herein are true and complete to the best of his/her knowledge; and (2) agreeing to accept the obligation to comply with NSF award terms and conditions if an award is made as a result of this application. Further, the applicant is hereby providing certifications regarding conflict of interest (when applicable), flood hazard insurance (when applicable), responsible conduct of research, and organizational support as set forth in the NSF Proposal & Award Policies & Procedures Guide (PAPPG). Willful provision of false information in this application and its supporting documents or in reports required under an ensuing award is a criminal offense (U. S. Code, Title 18, §1001).

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By electronically signing the Certification Pages, the Authorized Organizational Representative (or equivalent) is certifying that there is organizational support for the proposal as required by Section 526 of the America COMPETES Reauthorization Act of 2010. This support extends to the portion of the proposal developed to satisfy the Broader Impacts Review Criterion as well as the Intellectual Merit Review Criterion, and any additional review criteria specified in the solicitation. Organizational support will be made available, as described in the proposal, in order to address the broader impacts and intellectual merit activities to be undertaken.

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By electronically signing the certification pages, the Authorized Organizational Representative is certifying that the organization will be or is in compliance with all aspects of the United States Government Policy for Institutional Oversight of Life Sciences Dual Use Research of Concern.

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(This certification is only applicable to travel proposals)

By electronically signing the Cover Sheet, the AOR is certifying that prior to the proposer's participation in the meeting, the proposer will assure that the meeting organizer has a written policy or code-of-conduct that addresses sexual harassment, other forms of harassment, and sexual assault, and that includes clear and accessible means of reporting violations of the policy or code-of-conduct. The policy or code-of-conduct must address the method for making a complaint as well as how any complaints received during the meeting will be resolved. The proposer is not required to submit the meeting organizer's policy or code-of-conduct for review by NSF.

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(This certification is only applicable to career-life balance supplemental funding requests)

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AUTHORIZED ORGANIZATIONAL REPRESENTATIVE		SIGNATURE	DATE
NAME Angela Cocchiarella		Electronic Signature	Jul 25 2022 02:26 PM
TELEPHONE NUMBER 352-294-3972	EMAIL ADDRESS anjoli88@ufl.edu	FAX NUMBER	

Overview

Employers across the United States are currently facing a crisis of labor supply in the wake of the pandemic, driven by a shift in the factors influencing the prices at which people are willing to work [1-7]. Simultaneously, prospective homebuyers find themselves faced with rising costs and fierce competition for housing, with low-income, minority, aging, and veteran communities hit hardest by rising home prices [13-145]. Addressing these challenges is critical to economic recovery, and it requires researchers to supplement economic understanding of behavior with studies of psychological and cognitive mechanisms underlying pricing [10-22] and how they affect individuals' behavior in housing and labor markets. This proposal will iteratively build pricing models by extending approaches developed by the PI [18,19] to explain how people assign value in applied, dynamic, multi-attribute environments. It will then apply these models to predict behavior among active jobseekers and prospective homebuyers as part of ongoing financial education programs. Finally, it will compare graded judgments like price to decision-making in order to explore price-driven "nudges" to help people be more patient, honest, thorough, and less polarized [27-30].

Understanding behavior in complex pricing scenarios requires a base of empirical research that informs our models of pricing. To build this base, we will gather experimental data on 1) risky and intertemporal pricing (*How much would you pay now to avoid a 50% chance of losing \$100 in 3 weeks?*); 2) how people assign price in applied multi-attribute and multi-alternative scenarios (*How much would I bid for a house with 3 beds, 2 baths, etc.?*); 3) the sensitivity of prices to competition and future prospects (*Should I bid \$350k now or save for a better house?*); 4) people's ability to make and improve their judgments by making price assessments and bets (*How much would you bet that X will happen / that you are correct?*). Data from these experiments will inform models of pricing and individual differences in preference, creating new, accessible tools for understanding the psychological factors driving the crises of labor and housing markets.

A pivotal education goal of this proposal is to make pricing models – and the tools of cognitive modeling more generally – accessible to a wider audience. This will be done by developing simple online tools for model fitting using machine learning [34-38], integrating computational modeling into graduate and undergraduate education, and offering workshops to researchers seeking to implement these methods. In turn, the models will be integrated into financial education programs through a collaboration with the University of Florida's community education extension (Institute for Food and Agricultural Science / IFAS). This IFAS collaboration will also allow us to recruit participants who are active on the labor and housing markets, examining how performance on our tasks and models relates to real-world outcomes.

Intellectual merits

The proposed work would transform our empirical, methodological, and theoretical understanding of the cognitive mechanisms underlying pricing in several ways. First, the experimental results will help us understand how price is assigned across contexts and test model mechanisms related to rejection, losses, delays, anchoring, and attention. Second, the machine learning-based model fitting tools we develop will make computational modeling more accessible to researchers, generating new tools for future work on price, wage, and subjective value. Finally, the modeling approach will formalize our understanding of how subjective value is expressed across these contexts, leveraging advanced models and fitting tools approaches developed by the PI [18-19,23-24] to estimate meaningful individual differences in cognitive processes relevant to value, motivation, and satisfaction with pricing outcomes like wages and home prices.

Broader impacts

Outreach through the UF IFAS extension program will allow us to assess the efficacy of financial literacy education programs, evaluating how participants' risk and delay preferences predict behavior on labor and housing markets. The IFAS outreach program through which we run studies will guide future public policy on minimum wage, improve real estate projections, create public-facing educational resources to help jobseekers and homebuyers, and help improve financial education programs across the country. The proposal also directly enhances graduate and undergraduate education, developing quantitative research skills in students from interdisciplinary backgrounds by advancing machine learning approaches to modeling [34-38] that make our models accessible for researchers without modeling backgrounds or computational resources. These new approaches will be accompanied by conference-affiliated workshops and a week-long local workshop dedicated to providing modeling education to researchers, with a focus on promoting data science skills among participants from disadvantaged backgrounds.

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Overview of Research & Educational Objectives

In the wake of the pandemic, employers across the United States are currently facing a crisis of labor supply driven by a shift in the factors influencing workers' desired wages [1-3]. These employment shortages are particularly perilous for small, local, and minority-owned businesses [2,3], rural communities [4], and female and minority jobseekers who have borne the brunt of the economic impact of the pandemic [5,6]. Simultaneously, the price of housing and competition among homebuyers has risen dramatically such that millions are being priced out of their homes, overburdened by housing costs, facing eviction, and struggling to find a place to live [138,139]. The housing and employment crises are also disproportionately affecting vulnerable populations like communities of color [140,141], veterans [142], and older adults [143]. Although not entirely driven by individuals, these problems are fundamentally linked to people's changing priorities related to the subjective value of housing, work, and compensation [7], but unlike traditional problems in behavioral economics, they revolve around issues of *pricing* rather than issues of choice. People reverse their preferences depending on whether they must decide between options compared to when they must assign monetary values to each one [18,19,45,46], yet nearly all research on subjective value has focused on the former. Currently, a lack of scientific knowledge education on how people assign value (utility) in pricing situations like housing or job markets jeopardizes the economic livelihood of the United States by failing to link economic behavior to individual-level cognitive processes.

This proposal seeks to develop this understanding by bringing together cognitive and economic models of value to bear on pricing. On the economic side, we have comprehensive rational utility theories of how people should balance the value of leisure time with that of paid work (marginal rate of substitution between leisure and income) [10,11] and how factors like interest rates ought to affect home prices [153]. However, the present challenge of labor and housing prices pertains to the utilities of subjective, psychological features: how do people integrate all the features of a job or a house into a price they would be willing to accept or pay, and how do they grapple with issues like fairness in wages and competition with other employees or homebuyers? And once they succeed in finding a house or a job, how does the price influence their satisfaction? Understanding the psychology of utility and its relevance to labor and purchasing is central to areas of NSF focus like the Future of Work at the Human-Technology Frontier [12-17].

To address these challenges, psychological and economic models have naturally begun incorporating more and more elements of cognitive models [12-22], such as sequential sampling [20], attention / bounded rationality [12], response times [13], and process tracing [14]. However, despite a deep literature in both psychology and economics devoted to understanding the choices people make [15-17], there is much left to be understood about how people assign prices [18,19]. To address this gap, the primary research goal of this proposal is to create an accessible, dynamic cognitive model that explains how people assign prices in complex, multi-attribute, and applied contexts. We propose a series of experiments that will challenge these models to quantify the relationship between features of an item (e.g., a job) and the price people assign to it (e.g., desired wage) by extending models developed by the PI [18,19]. These will advance our empirical understanding of pricing in complex and realistic environments, accomplishing four objectives:

Research Objective #1: Model the dynamic cognitive processes that underlie pricing in complex environments. This will be achieved through experiments looking at multi-attribute pricing, including eye-tracking data to evaluate what information people consider (Cluster #1). We will adapt existing models of price created by the PI [18,19] to construct a cognitive model of pricing behavior that integrates responses, eye fixations [20-22], and response times [23-25] to form a complete picture of how people solve challenging price problems.

Research Objective #2: To create simple, accessible tools for applying these models to real data, including online tools for assessing risk and delay preferences. One downside of dynamic models of price is that they rely on simulation and tools like approximate Bayesian computation to fit to real data. To make these models more accessible and usable for researchers, we have created machine learning tools for model fitting that instantly map observed price data onto individual difference parameters quantifying risk and delay preferences. These tools will be integrated into a software package that researchers can use to fit the complex pricing models we develop (Cluster #2).

Research Objective #3: Apply these models to understand pricing in realistic environments, including the price of labor / reservation wages during the job search. Through a collaboration with

UF IFAS Extensions, we will collect data from prospective homebuyers and jobseekers across different economic sectors and geographic locations (Cluster #3) about what factors they (plan to) prioritize when buying a home or looking for employment, run simulated house hunting and job search experiments, and relate this information to predict how these factors predict their search in real applied settings, relating patterns of behavior on experimental tasks to financial education program participants' real behavior on the labor and housing markets.

Research Objective #4: Identify strategies for pricing that reduce or eliminate biases induced by choice. Based on our model of how people assign price, we expect that price-oriented strategies can make people less sensitive to bias due to context [27] become less polarized [28] favor long-term outcomes over short-term payoffs [19], and make better metacognitive assessments [29]. In Cluster #4, we will test the efficacy of price-oriented manipulations to create “nudges” [30] that reduce the biases inherent to human choice and help people focus on long-term financial goals.

The model-based analyses of these data are also central to the educational component of this proposal. A major hurdle in psychology research is equipping researchers with the right tools for constructing and testing theories. Computational modeling offers a solution to this problem, but it has historically been consigned to specialized subfields of psychology due to the statistical and programming skillset these models require [31-33]. To address this problem, we will create machine learning methods for model fitting that make them easy for anyone to use. The results of the empirical studies and modeling endeavors will then be used to improve financial education programs. This addresses three education objectives:

Education Objective #1: To make the tools of cognitive modeling accessible to researchers.

Computational models will be made more accessible by using a recently developed machine learning approach [34-38] that allows a modeler to create neural networks that instantly map observed behavioral data onto best-fit model parameter estimates. Using this approach, we will create simple online modeling tools that allow researchers to upload pricing data and receive parameter estimates describing their participants' preferences for (e.g.) risk taking, delay discounting, and price biases.

Education Objective #2: To integrate modeling insights into financial education programs.

The modeling approach we develop under Research Objective #2 will be integrated into online behavioral assessments that gather participants' responses on a few pricing problems and use the resulting data to make inferences about their risk and delay preferences. These assessments will then be integrated into financial education programs through a collaboration with faculty in the UF Institute of Food and Agricultural [IFAS] Extensions programs. Following participants in the First Time Homebuyers and Money Matters financial education programs, we will identify properties of pricing strategies that lead to satisfactory wages and home bids, provide individualized feedback about risk and delay preferences that promote good savings strategies (e.g., focus on *outcomes* / *pricing* when thinking about saving), and evaluate the impacts of financial education programs on wage and housing outcomes.

Education Objective #3: To create educational materials on our empirical results and modeling approaches.

Research Objectives #2-3 evaluate how new workplace risks, job features, changes to unemployment benefits, and shifts in minimum vs reservation wages could affect the compensation that people demand for their labor. We will create public-facing educational IFAS documents (as in [64,65]) that are used in outreach programs, making these results accessible to prospective homebuyers and jobseekers [1-3]. Furthermore, the modeling approach we develop is useful for a variety of models and fields beyond psychology and economics – to make them more widely usable, we will host local and conference-affiliated workshops on machine learning approaches to computational modeling.

Expected Significance & Intellectual Merit

Static, deterministic theories like expected utility and prospect theory have formed the foundation of our understanding of subjective value [39-44]. However, pricing and decision-making are governed by *dynamic* cognitive processes that can only be understood with appropriate generative models [18,19,24]. The proposed efforts would potentially transform our understanding of multi-attribute pricing, wage demands, home bids, and the complex dynamics of pricing in competitive markets. In doing so, we would create free online tools for model parameter estimation that allow anyone to model their pricing data. Our proposed

research will further the science of pricing and decision-making in three ways. First, it will pursue an empirical and model-based understanding of how people assign prices and how these prices change with attributes of jobs and homes (Research Cluster #1). Second, we will create new modeling tools for automatically assessing latent preferences for different components of value, including risks, delays, and gains / losses (Cluster #2). Third, these approaches will be used to understand current labor and housing markets by following participants in financial education programs aimed at helping them find jobs [Money Matters] and buy homes [First Time Homebuyers] (Cluster #3). Finally, the proposed work will examine how the cognitive processes underlying pricing can be manipulated to promote good strategies for assigning value (Cluster #4). Put together, this proposal will provide empirical, methodological, and theoretical advances in our understanding of pricing and subjective value while informing and promoting financial education. It will broaden our equip researchers with advanced modeling tools, create methods for accessible cognitive modeling, and develop novel behavioral interventions that enhance information search, improve metacognition, and promote long-term financial planning.

Research plan

Preliminary / motivating results. The research we propose below is motivated largely by gaps in existing theories of pricing. Among the most important problems with models of subjective value is their ability to handle *preference reversals*, where the method of eliciting value determines the apparent preferences that people express. In risky choice, it has been known since at least the work of Lichtenstein & Slovic [45,46] that people tend to favor safe options (e.g., 90% chance of \$4) over risky ones (e.g., 20% chance of \$20) in choices, but assign a higher price to the same risky option when presented separately from the safe one. Our recent results have also shown that participants choose a smaller, sooner option (\$10 today) over a comparable larger, later option (\$15 in 6 weeks) in binary choice, but assign a higher bidding price to the larger later option than to the smaller sooner option. Preference reversals like these illustrate the need for theories specifically of pricing, as opposed to theories that are merely adapted from decision making: no matter how much decision data we collect, we will never be able to use models of choice alone to fully predict people's behavior on pricing problems, including wages or housing bids.

Beyond preference reversals, our recent results have further justified the need for more complete theories of pricing. Our work in risky choice [18] as well as intertemporal choice [19] has shown that price is influenced heavily by when it is elicited. For example, people exhibit much strong biases based on their perspective as a buyer (employer) or seller (laborer) that are exacerbated by time pressure. Furthermore, our work has shown that prices are often heavily skewed (Figure 1), reflecting complex interactions between the attributes of items (such as jobs) and the prices people assign to them. As a result, classical static theories like expected utility and prospect theory do extremely poorly in accounting for these patterns of results compared to dynamic price accumulation models we have developed [18].

Put together, these results emphasize the need for dynamic, stochastic models of pricing that can explain why preferences measured in monetary value appear to diverge from choice. This problem extends beyond models of price in risky decision-making contexts. Putting theories of decision-making and pricing back together into a unified (dynamic, stochastic) theory of subjective value is fundamental to tackling more complex problems like wage and home pricing. The research we propose will develop models that consider the distributions of prices people assign to possible options, the time it takes to make them, and examine how people allocate attention during multi-attribute pricing problems, allowing us to compare pricing and decision-making in the kinds of multi-attribute contexts that jobseekers and homebuyers encounter. Research Cluster #1 will leverage process-level data like response times and visual fixations to create these models of subjective value that can account for the processes underlying pricing.

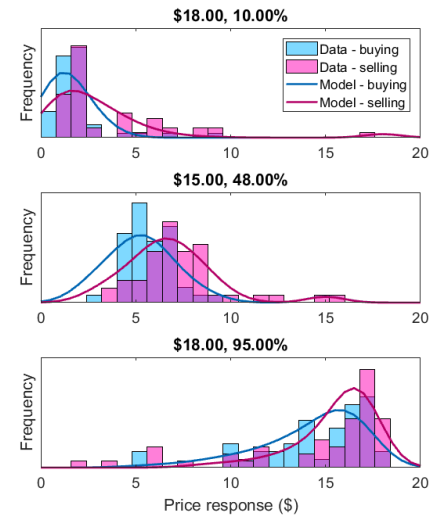


Figure 1: Distributions of prices assigned to risky prospects. Price distributions are often heavily skewed, reflecting underlying dynamic cognitive processes that are sensitive to features of the price objects.

Research Cluster #2 will take these models and make them accessible. Using machine learning approaches, we will create neural networks that can directly map observed data onto process-level model parameters. An example of this type of implementation is shown in **Figure 2**, where we have taken simulated data from 20 example gambles (15, 35, 65, or 85% chance of \$4-20) and trained a deep neural network to map responses directly onto the parameters that generated them. These methods for parameter estimation are highly accurate and grant insights into processes related to utilities (e.g., the value of wages), starting point biases (reservation wage), and thresholds (e.g., aspirations). To test this approach to parameter estimation and set its priors, we have gathered a preliminary sample of 300 participants. The distributions of parameter estimates correspond very strongly (average $r = .86$ across parameters) with estimates obtained via traditional maximum likelihood estimation with simulated likelihoods / probability

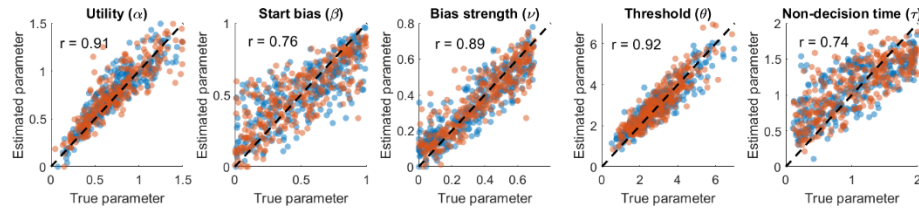


Figure 2: Performance of a machine learning approach to parameter estimation of our pricing model [18] using data from 20 different risky pricing problems (\$4-20, 15-85%). Each plot shows the true value of a parameter of the model (x) against its estimated value generated from a neural network trained on simulated data from the model (y). Predictions from the training set are shown in blue, while out-of-sample tests are shown in orange.

Research Cluster #3 extends our work to pricing in complex environments like wage-setting to address wage and home price issues by examining the pricing strategies that are implemented by current jobseekers and prospective homebuyers. We will study these by directly recruiting local (Florida) jobseekers and homebuyers through ongoing financial education and outreach programs, assessing their behavior in on experimental pricing tasks, and then connecting their behavior on the experiments (quantified by our modeling approach) to their real-world job and home search outcomes. This will allow us to evaluate what strategies and individual differences predict success on real pricing problems.

Finally, although the diverging patterns of preference between price and choice are problematic from the standpoint of theory building, they offer an unusual opportunity to create interventions that help people make better judgments. The final cluster of research we propose (Cluster #4) will create interventions / “nudges” [30] leveraging pricing to help people make better judgments [29], consider more information more carefully [28], and achieve long-term savings goals [27].

Sample size justification. For most of the studies we outline below, a primary or secondary goal is to connect latent traits or measures (self-report / behavior measures) or understand the effects of manipulations. To maximize the reliability and predictive validity of our measures, we use hierarchical Bayesian methods for model fitting [24,25], which allow for conclusions for or against null hypotheses. Power analyses are aimed at identifying how much data is needed to conclude in favor of the null hypothesis when it is true [26]. This analysis is shown in **Figure 3**. We typically want to be able to conclude in favor of a null hypothesis (Bayes factor > 3) at least 80% of the time when there is a true null correlation, which requires approximately 85 participants. In other cases, we want to be able to conclude in favor of a null hypothesis at least 95% of the time, which takes approximately 300 participants. This is used to guide sample size estimates in the experiments outlined below.

Participants in each of our studies will be compensated for their time and receive an incentive-compatible bonus. For example, participants would exchange a fixed sum for a risky payoff if their bid is high enough in a

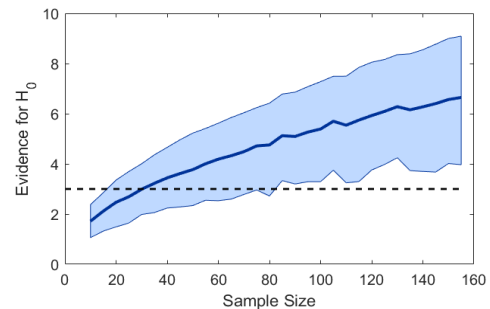


Figure 3: Power analysis for the relationship between two parameters in a Bayesian analysis. Blue line represents the expected evidence (Bayes factor) for a null effect, and the shaded region describes the range containing the 20-80th percentiles of Bayes factors.

lottery-buying condition or be paid based on the decisions or judgments they make (see [18] for a complete payment scheme), the rate at which their bids are accepted, or their overall performance in terms of the accuracy of their decisions and probability estimates (Cluster #4 / Study 2).

Modeling. The theory of subjective value we have developed is central to the experiments and several of the outreach efforts proposed here. Our model of pricing on the tasks will be based on the PI's continuous, dynamic Price Accumulation model [18,19], shown in **Figure 4**. In this model, a person makes

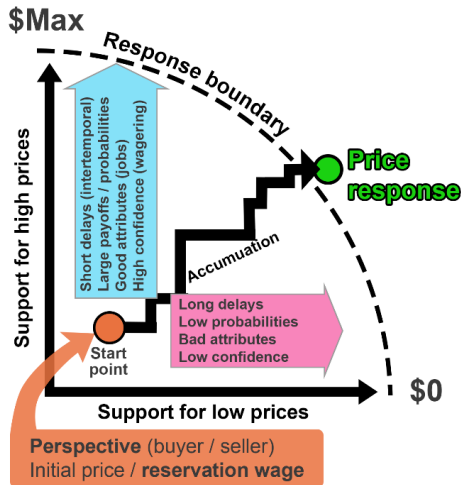


Figure 4: Overview of the modeling approach. Participants' initial state (affected by, e.g., buyer-seller perspective or reservation wage) is adjusted as they consider attributes of the price object, until one price receives enough support to be selected (corresponding to their state crossing the response boundary).

a pricing judgment by integrating an initial state (based on some initial biases related to a participant's "default" reservation wage, or the asking price on a house) with information about attributes of a job. Positive attributes like large, certain, imminent outcomes drive them toward higher prices (up), while negative attributes like small / negative, risky, or delayed outcomes drive them toward lower prices (right) based on how heavily they are weighted. It has five main parameters: an initial start point bias (what price someone starts at), a start bias strength (how influenced they are by their initial price), a threshold (indicating how certain they wish to be before assigning a final price, controlling the speed-precision trade-off), a utility parameter or a discounting rate (indicating how similar dollar values or delays are represented relative to one another), and a non-decision time parameter (indicating the length of time to encode stimuli and enter responses). Put together, these parameter estimates allow us to quantify the degree to which participants are influenced by anchors (e.g., asking prices), how their representations of subjective value change over time (and thus change under time pressure), how quickly they can assign prices (critical for competitive markets), and how risk preferences affect price judgments. Naturally, our model will need to be extended to account for behavior in new

scenarios like wage-setting, wagering, shifting options, competition, and metacognition. We expect to test three new modifications to the model over the course of the experiments we outline below:

- **Model addition #1: Attribute / attention weights** – as in many dynamic models of similarity, categorization, and utility, each component of an option (e.g., number of bedrooms / bathrooms, location, and square footage for houses) must be assigned a relative importance [156,157]. We expect to find differences in attribute weights across participants that correspond to priorities for different features of jobs and houses.
- **Model addition #2: Multiple anchors** – Our recent work on models of continuous responding has indicated that judgments like price can be influenced by the presence of multiple anchors [148]. For example, we should expect that housing prices are influenced by not only the asking price but also the buyer's budget, the price of their current house, and the prices of other houses. Likewise, a desired wage may be influenced by aspiration level, current wage, and the posted or offered wages for a position. We will integrate these into the model using multiple-anchor accumulation theory [MAAT] to drive accumulation, a dynamic anchor-based model developed by the PI.
- **Model addition #3: Rejection thresholds** – A major component of finding the right job or house is knowing when to forego an option. For options where participants are not interested in bidding or setting a wage, the model needs some mechanism for eliminating the option from consideration. To do so, we will introduce "negative" choice boundaries that allow options to be eliminated when their attributes are seen too negatively (described in work by the PI [67]).

In the eye tracking study, attribute weights will be loaded onto fixation time and scaled based on participants' ratings of the value of each attribute, so that time spent looking at negative attributes drives accumulation toward low bid prices (or high wage demands) and time spent looking at positive attributes drives accumulation toward high bid prices (low wage demands). This model already accounts well for patterns of price in both risky and intertemporal choice [18,19], so we expect it to be straightforward to

extend it more complex stimuli as long as they can be quantified in terms of negative and positive features. The eye tracking data will further improve our ability to predict what price participants will assign to different prospects by accounting for trial-level variation in what they pay attention to [20-22,152]. To ensure that our models are usable by other researchers, they will all be implemented in Python, R, or MATLAB and the analysis and modeling code will be uploaded to Open Science Framework project pages.

Research Cluster #1: Multi-attribute pricing

As we suggested in the preliminary results, static models of pricing like expected utility or prospect theory [15,39-43] miss major empirical results because they do not account for the influence of time or the shape of distributions of prices [18,48] (**Figure 1**). Conversely, dynamic models of pricing that can handle these effects [18,19,49] have focused on simple cases where decision makers only need to consider a single payoff and a single probability (How much would you pay / accept for a 50% chance of \$10?). As a result, there is no model of price that accounts for the dynamics of attention and how they contribute to pricing as a participant considers multiple features of an item. The first cluster of studies on this proposal focuses on addressing this problem, using multi-attribute pricing problems complemented by eye tracking to create a more complete model of pricing.

Study 1 – Risky intertemporal pricing. We have already collected data looking at the combination of three most common components of economic decision making into a dynamic multi-attribute model of pricing: gains/losses, risks, and delays. An outline of this experiment is shown in the upper panel of **Figure 5**. Participants were shown a risky intertemporal prospect [50] where they could receive one of multiple potential delayed rewards (e.g., a 30% chance of \$10 in 5 days, 70% chance of \$5 in 35 days) and be asked how much they would pay for that prospect or how much they would sell that prospect if they had it (to assess buyer-seller differences [51]). To connect behavior between choice and pricing conditions, and relate performance to self-report measures of risk-taking and impulsivity [134-136], we recruited 12 participants who each returned for 6 sessions each, completing choice, buying, and selling conditions which were crossed with time pressure (speed emphasis / precision emphasis). This allows us a sufficient number of trials per condition to fit the model and to assess the test-retest reliability of our behavioral measures and model parameters. The data will be analyzed using a model that includes attribute weights (Model addition #1), allowing us to estimate the relative importance of payoff, delay, and risk for each individual participant, the impact of these parameters on price and choice, the degree to which the effects of anchors (payoffs) is attenuated by the presence of delays and risks, and how attribute sampling and pricing (relative to choice or risk/delay alone) are affected by time pressure.

Study 2 – Evaluating job and house features. Study 1 should provide experimental evidence to constrain a dynamic, multi-attribute model of risky intertemporal choice. However, many real-world pricing scenarios like wage-setting and homebuying do not lay out outcomes clearly in terms of risks, payoffs, and the time at which they will be realized. The second study will draw a connection between the aspects of real jobs and homes that are most important to our participants and how these aspects are considered by participants when evaluating their overall value. To do so, we will compile a list of attributes for (1) homes, and (2) positions in twelve different job sectors: retail, agriculture, manufacturing, health care, construction, education, real estate, business services, information, finance, hospitality, and government. Attributes will be compiled from a combination of ads scraped from job sites and information provided on house hunting sites like Zillow and open MLS systems.

We will tabulate the 10 most-mentioned attributes of houses and jobs in each sector across our participants and scraped descriptions. Next, we will evaluate the benefits, costs, risks, and timeline of these job attributes. A second set of participants will rate for each attribute (e.g., positive social environment) how positive / negative these outcomes are, how persuasive that attribute is, how likely they believe they are to

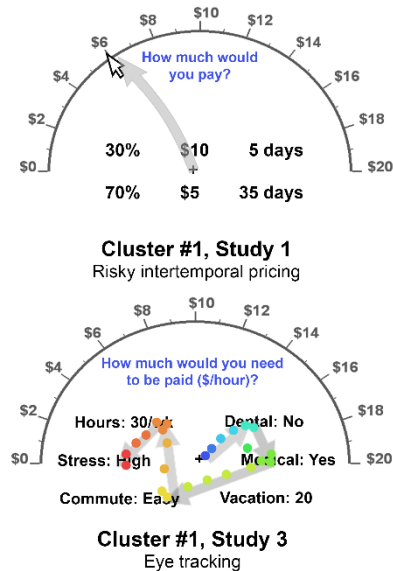


Figure 5: Diagram of Study 1 and Study 3 of Cluster #1. The prices, response times, and visual fixations from these tasks will inform our theories of pricing.

encounter such an attribute, and how long it would take to realize the benefits of each attribute. For example, a good pension might be rated as highly desirable, somewhat persuasive, common in potential jobs, and temporally distant. To ensure that these assessments are calibrated to those of typical U.S. residents, we will recruit a demographically representative set of 300 (minimum required for a representative sample) participants through Prolific Academic to supply these ratings. These ratings and estimates will allow us to connect the results of Study 1 (risky intertemporal choice) more directly to those of Study 3 (pricing jobs and houses) and allow us to better extrapolate results to the general U.S. population.

Study 3 – Eye tracking. Although we might ordinarily expect jobseekers and homebuyers to weigh every aspect of a position and its utility [39-43], people often rely on a limited subset of the available information (in a job ad) to make their judgments [52-56]. Heuristics for decision-making are pervasive, yet few heuristics for making continuous judgments like pricing have been identified (with the exception of the risk-reward heuristic [57]). To truly understand the pricing process, we must identify the order in which people search for information, what information they attend or ignore, and how different attributes determine price. Therefore, we draw upon process-level data (response times and visual fixations) to identify any heuristic sampling strategies that participants use to simplify the pricing process, and contrast these against choice heuristics they use to make decisions between jobs or houses.

Theories of choice have been informed by eye tracking that examines how attention is allocated during the valuation process, suggesting that gaze allocation determines the weight assigned to each attribute [20-22,44,58-61]. Study 3 will implement eye tracking methods to understand how gaze is allocated as a window onto how attention drives the pricing process. This study is outlined in the bottom panel of **Figure 5**. Participants will view a display of 4, 10, or 20 attributes (**Figure 5** displays 6 for visual simplicity) of houses or jobs in each sector, each of which will vary in quality (e.g., low to high physical toll, for manufacturing jobs). Attributes will be consistently placed on the screen so that participants know where to look for each one. Participants will be permitted to view as many or as few attributes as they like, before clicking the mouse to indicate they are ready to enter a wage demand or a house bidding price using the radial scale shown. Visual fixations will be recorded during the task using an eye tracker and used in the model to predict the price participants assign to each job and the time it takes them to determine the price, as in our previous modeling work incorporating eye tracking [44]. This will be integrated with ratings from each participant of the goodness / badness of different values of each attribute [see 62], which will be collected prior to the start of the eye tracking study. To examine relationships between behavior across conditions, we expect to recruit 85 participants (2 sessions each) for this study, who will each complete both the job and the housing conditions as well as the different numbers of attribute conditions (4, 10, or 20). We expect that participants will show greater heuristic use in the more complex / greater number of attribute conditions, corresponding to fewer visual fixations per attribute and more attention weights with value of zero; higher thresholds with a greater number of attributes, producing greater response times; and a weaker relationship between the “true” (attribute-weighted) value of the option and the assigned price.

Cluster #2: Accessible tools for model fitting

Despite the power of computational modeling approaches to studying behavior, researchers are constrained by time (training, fitting) and resources (supercomputers, expert knowledge) on their use [31-33]. A unique strength of our lab is the capacity to make complex computational modeling approaches simple and accessible for researchers without advanced training in these areas [e.g., 18,19,28,68]. Machine learning tools for model fitting afford us the opportunity to use an approach called *amortized inference* [34-38], where a modeler carries the burden of parameter estimation by pre-training a neural network to map a set of observed data (in this case, prices; top of **Figure 6**) onto the parameters of the price model that was used to generate the data (bottom of **Figure 6**). This is done by simulating a large volume of data from the model (e.g., 1 million data sets) across a variety of parameter values (drawn from priors), then using the simulated data to teach the network the functional relationship between data and model parameters. Once trained, this network can be deployed on real data to instantaneously estimate best-fit model parameters from raw (price) data [34], characterizing individual differences in the utilities people assign to different outcomes, their starting points or initial biases for pricing (such as base reservation wage), and the amount of consideration they give to a particular price object (thresholds; parameters shown in **Figure 2**).

Research Cluster #2 is aimed at vetting an amortized inference model of pricing and making it robust to data uploaded by users (Study 1) and then integrating a model into an online experiment that provides financial education to participants (Study 2). To accomplish the latter, we will collaborate with the University of Florida's Institute for Food and Agricultural Science [IFAS] Extension. The IFAS Extension holds classes and special programs in county offices statewide, granting access to jobseekers and first-time homebuyers in rural, urban, minority, majority Spanish-speaking, and low-income communities.

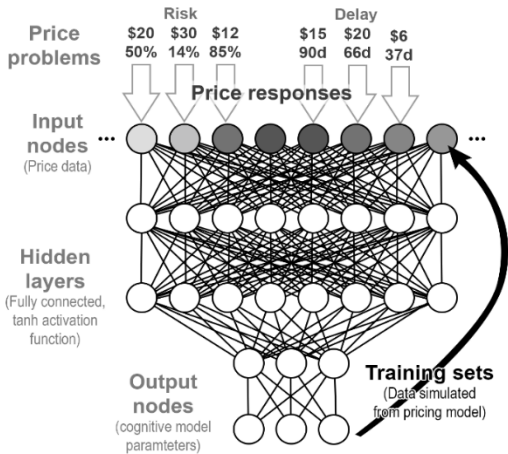


Figure 6: Outline of amortized inference approach to creating accessible models. Simulated data (training sets) are used to teach a neural network the relationship between behavioral data on a price task (top) and corresponding cognitive model parameters (bottom). The results of a network we have trained are shown in **Figure 3**.

As part of the IFAS Extension services, the research materials will be translated into Spanish to serve a more diverse community of participants. The IFAS Extension collaborators run a First Time Homebuyer's Class and Money Matters course in both English and Spanish. Participants in these two programs currently benefit from online resources including articles, videos, and worksheets developed by IFAS staff. We will expand this program by developing online assessment tools for risk and delay preferences, integrating data collection in Reach Cluster #2 (Outreach & Education #2).

Study 1 – Parameter estimation tools for researchers. The first goal of our machine learning approach to parameter estimation is to create accessible, usable tools for researchers to use pricing models. The present version of the network, shown in **Figure 2** and **Figure 6**, uses a fixed set of stimuli to ensure a consistent mapping between responses (prices and associated response times) and parameters / posterior error [35].

Realistic experiments for which users (researchers) upload data will feature a range of sample sizes, stimuli, and participant responses. Our goal in Study 1 is to create a freely available online tool that can deal with these variations in study design. This will be done by passing the network not only the price and response time for each problem (gamble or delayed prospect for risky or intertemporal choice, respectively) but also the stimulus itself. This will create a 4xN input structure (flattened to 4*Nx1), where there is a payoff, delay/probability, price response, and response time for each of N stimuli in the experimental data. To address varying values of N, we will leverage a combination of multiple neural networks trained on different sample sizes [70], multiple imputation of values in the first hidden layer of the network (when there is too little data in the input) [63] and bootstrapping across multiple subsets of the data to obtain best estimates (when there is too much input data).

This approach will be tested using multiple simulation studies (testing a wide variety of input samples and stimuli) and applied to human data gathered from the UF Department of Psychology participant pool. We will collect 300 trials (maximum number of data points the network will be designed to accept) of price data from each of 85 participants [77], then use subsets of the real data (e.g., 55 random trials) to test the neural networks. We will evaluate each of the networks, hidden layer imputation, and resampling / bootstrapping methods against cutting-edge model fitting methods for different sample sizes and participants, iteratively adjusting each component until the networks can achieve at least a correlation of $r = .8$ with the hierarchical Bayesian maximum a posteriori estimates. This will ensure that the machine learning approach to parameter estimation agrees with our best parameter estimation approaches and are robust to different experimental designs. The networks will then be integrated into online tools (server-side Python) so that they can be used to generate parameter estimates from data uploaded by researchers. This will allow users to obtain model parameters with no need for the computational and statistical burdens imposed by parameter estimation, creating equity in research by making the most powerful modeling tools usable for researchers without access to advanced training or computational resources.

Study 2 – Online assessment of risk and delay preferences. The second study in Cluster #2 will use the trained networks to complement online experiments assessing risk and delay preferences via pricing problems. In the risky choice experiment, we gather pricing data from participants on 20 risky pricing problems ("What is the maximum amount you would pay for a 35% chance of \$20?"). In the delayed choice

experiment, these are replaced with 20 delay problems (“What is the maximum amount you would invest now for a guaranteed \$20 in 90 days?”). These will be crossed with a gain/loss manipulation (“How much would you pay to **avoid** a 35% chance of **losing** \$20?”). The experiment takes only 5-10 minutes in total. The data that participants provide will be used to estimate model parameters related to risk aversion, loss aversion, delay discounting, thresholds, start biases, and non-decision times.

As before, this trained network will be built into the server-side Python code that executes the experiment, allowing us to return model parameter estimates to participants in the online experiment immediately following completion. With each participant who takes the experiment, this database will be updated so that it can constantly improve the feedback it gives to participants. Once a participant has completed the study, they will receive feedback about their estimated parameters, along with an explanation of what the parameters mean and the quantile of their response relative to other participants (e.g., “You are in the 75% percentile of the risk-taking parameter, indicating that you are more likely to take a chance on a small chance of a big payoff than most participants.”). For participants in Money Matters and First Time Homebuyer programs, they will also be debriefed about how these preferences relate to job or house hunting, respectively. They will also be pointed to the IFAS Money Matters educational resources [64,65] and (once prepared) the public-facing documents we produce on wage pricing and decision-making. Because this study will double as an educational tool, it is intended to stay up in perpetuity, but we anticipate disseminating the results by the end of Year 5 (outline in **Figure 8**).

Cluster #3: Studies of real house hunters and jobseekers

The goal of Cluster #3 is to push the basic research findings gathered under Cluster #1 into an applied setting where our findings can be leveraged to understand the labor supply and home pricing problems the nation is facing. We will gather experimental data from jobseekers and homebuyers across the state of Florida about their wage demands or home bid prices (Study 1), then follow up with these participants at one month and one year later to understand how their performance on the task predicted the wages they obtained or the home that they bought (or whether they succeeded at all in finding a job / house; Study 2).

Study 1 – Behavior on laboratory pricing tasks. The first study will focus on evaluating Florida jobseekers’ and homebuyers’ behavior on pricing tasks. We will recruit 300 participants from the Money Matters program and 300 participants from the First Time Homebuyer program, recruiting through the in-person local IFAS offices in each Florida county and the online (Zoom) classes, aiming for at least five participants from each county for geographical diversity – in particular, we want to examine any urban-rural divides in wage and home pricing. Jobseeking participants will be asked a series of basic questions about what type of jobs they are looking for (if any), what sector they want to work in, length of unemployment / time in their current position, what concerns they have about (returning to) working, whether they are taking unemployment benefits, and how much they would need to be paid to work a “typical” job in their favored sector. House hunting participants will be asked questions about their housing budget, current housing situation (rent), desired house attributes (# bedrooms, bathrooms, sq. ft., location), their timeline for buying a house, and their expectations about the number of houses they will need to see and their local market.

Following these introductory questions, participants will complete an online experiment – coded in JavaScript – where they complete a simulated job search or house search as appropriate to the program in which they are enrolled. In this task, participants will be presented with a series of jobs or houses that appear and disappear sequentially on the screen, as shown in **Figure 7**. A highlight of each job or house will be provided on this screen based on what participants report as their most important considerations (e.g., stress, benefits, social environment, and meaningfulness of the work). As the trial progresses, participants can left-click to inspect a house or job and see more attributes (10, then 20 if desired) and enter a bid / desired wage, or right-click to remove that option from the screen. This allows us to test the rejection threshold component of the model (Model addition #3) and identify how participants are looking for positions in both a positive way (estimating price) and how they are eliminating options.

The stimuli will be calibrated to the budget that participants have (for homebuying) or the sector that they want to work in (for jobseekers) based using the information generated in Research Cluster #1, Study 2. The aim is to provide a challenging but realistic mix of good and bad options, making it difficult for participants to stop and assign a desired wage / bid to one of the options. If a participant’s wage demands are too high or house bid too low (relative to those assigned by participants in Research Cluster #1), the

request will be rejected, and they will continue the trial. If they enter a price that is accepted, the trial will finish and the participant will receive a monetary reward proportional to a weighted sum of the attribute values and their bid / wage demand (according to the participant's self-reported attribute importance assessed prior to the start of the experiment) to incentivize good performance; failing to find a job or house after 30 options are shown will result in no compensation for that trial.

One of the reasons that opportunities arise or disappear over time is the presence of competitors in the market who are willing to make more attractive or timely offers [69]. Despite high unemployment rates, the widely publicized labor shortages in the United States could make jobseekers more willing to wait for higher wages or better prospects (culminating in the “Great Resignation”), turning a perceived labor shortage into a real shift in its marginal resource cost [10]; likewise, high home prices are driven by fiercely competitive markets. To study this element of pricing, we will manipulate the competitiveness of the “market” in each experiment, with either high competition (jobs / houses disappear quickly, and wage / bids are rejected often) or low competition (jobs / houses disappear slowly, and wages / bids are rarely rejected). It has been shown that perceived competition can dramatically affect the amount of information people sample and how quickly they choose [71,72], but basic research on the effects of competition on pricing behavior is scarce. We expect that participants will have lower thresholds, lower start biases, a weaker anchor from their budget / reservation wage (Model addition #2), stronger anchors from other houses / jobs, sample less information, and assign lower wages (make higher housing bids) in the high competition condition. As with Cluster #1 (Study 3), we will use eye tracking during this experiment to examine what information participants attend, and use this information to inform the cognitive model of performance on the task. We expect a typical trial to take up to 1 minute each, for approximately 60 trials per session (30 / condition).

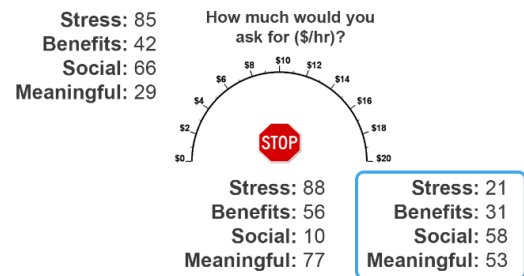


Figure 7: Illustration of the structure of the pricing task with shifting options. Options appear and disappear around the central location; participants enter a price by clicking on an option to see more attributes, or remove it from the screen so that another job / house can appear by right-clicking on it.

Study 2 – Follow-up surveys of jobseekers’ and homebuyers’ experiences. The second study in Cluster #3 aims to understand the connection between behavior on the laboratory task and participants’ actual performance on the labor or housing market. This study will focus on follow-up surveys with participants recruited for Study 1, aiming to understand the factors that governed their wage or housing prices. For two years after they take the initial experiment, participants will be prompted complete a monthly survey asking about different aspects of their job or home search experience. They will receive an additional \$5 for every time they complete the monthly follow-up survey, which should take less than 10 minutes.

These surveys will ask several types of questions: (1) simple questions about whether they have found a job / house and at what price; (2) if they found one, how long it took them to do so and how many they considered before finding one; (3) questions about the attributes of houses or jobs they have considered over the past month; (4) questions about their perception of the competitiveness of the market they are in; and (5) follow-up questions about how satisfied they are with their wage / purchase price of their home, if they did find a job / house since taking the experiment; and (6) model oriented questions about how strongly they considered their budget or reservation wage [start point bias, multiple anchors], how much those influenced their bids / wage demands [start point strength], what attributes wound up being most important in their considerations [attribute weights], how many jobs / houses they passed over and why [rejection thresholds], and how much time pressure they had to find a job / house [threshold]. We expect that model parameters in the initial laboratory experiment will predict patterns of behavior on the job and housing market, which will interact with the competitiveness of the market to predict their degree of success and the wages / bids that they price. Two key phenomena we are interested in are (1) if there is a “paradox of choice” where people are more satisfied with their jobs / houses in more competitive markets when there are fewer options available, and (2) if there is an effort/cost effect where people are more satisfied with houses they pay more to acquire [154,155].

Cluster #4: Nudges with pricing

Research on decision-making has unearthed multiple biases that accompany the cognitive processes involved in choice [11-17], yet pricing tasks are known to reverse the preferences people exhibit relative to decisions [45-49]. In some domains, this has the potential to yield better outcomes: our past work has shown that pricing results in reduced impulsivity relative to intertemporal choice [19] and that graded judgments like price can result in deeper and more representative information search, even eliminating elements of polarization and extremism that accompany the decisions. The final cluster of studies seeks to treat pricing as an assessment goal (judging value) and compare it to choice (discrimination goals) as a paradigm for forming beliefs. We examine the effect of having participants assign prices rather than make decisions in three areas where biases are prevalent but potentially malleable: information search, metacognition / confidence, and non-monetary intertemporal choice.

Study 1 – Information search. The first study in this cluster will examine how participants direct their search for information when they are tasked with pricing consumer products or deciding between them. Our previous work has shown that decision goals can create “polarized” information distributions, where extreme information (very good or bad attributes) is heavily weighted in choices while moderate information is severely underweighted [74,75]. Task directions aimed at estimating or assessing the difference between options rather than deciding between them appear to reduce this information polarization [28]. Based on these findings, we expect that a manipulation where participants must identify a price difference (“How much more/less would you pay for Option A than Option B?”) will be able to reduce opinion polarization and increase information sampling in consumer choice. If successful, this pricing manipulation would help people make better purchasing decisions by first assigning a value to their options. In this study, we will present participants with sets of 2-3 products and ask them to either choose their favorite (choice condition), report how much they would pay for each one (separate pricing condition), or report how much they would pay to receive one over another option (joint pricing condition). Behavior on these tasks will be related to self-report measures like dogmatism and need for cognition. We anticipate that participants will draw more information in both pricing conditions, and that this will be related to both dogmatism and need for closure [28,76]. We expect to require 85 participants to ensure sufficient power to detect these / null correlations.

Study 2 – Impulsivity. Our recent work has illustrated that asking people to assign a price to delayed prospects results in severely reduced delay discounting and a greater focus on delayed outcomes when options are priced individually [19]. To be a practical “nudge,” however, this approach should be able to manipulate preferences even in the presence of multiple options. To validate pricing as a manipulation to reduce impulsivity, we will present multiple smaller, sooner options and larger, later ones and ask participants to price each one. This will be compared against decision-making between the same stimuli. If pricing focuses attention on the outcomes, as our theories suggest [18,47,81], we should expect reduced impulsivity in pricing. We will examine this manipulation in simple intertemporal choice (e.g., \$10 today vs \$15 in 7 weeks) as well as decisions / prices in applied domains (e.g., drink alcohol now or maintain sobriety through the month; accept economic loss now or climate change consequences later) [82-87]. In each domain we will examine choice and prices for 40 choice pairs for a total of 160 trials (choice/price x basic/applied domain). We would expect a successful pricing nudge to orient preferences toward the long-term option across both basic and applied intertemporal choice, evidenced by higher prices assigned larger-later options than shorter-sooner options relative to the rate at which the larger-later is chosen over short-term outcomes in decisions. Manipulations will be carried out within-subjects with 85 participants [134,135], to ensure sufficient power to detect correlations between behavior and self-reports (see power analysis). Put together, Cluster #4 will create concrete strategies for improving information search, assessing people’s “true” underlying beliefs using wagering, and reducing impulsivity in when facing delayed consequences.

Outreach & Education Plan

The goal of education and outreach in this proposal is to promote the use of cognitive modeling in while leveraging research results to improve financial education and inform public policy. We focus on four areas where we can maximize the impact of our research results and expertise:

1. Creation of accessible online tools for model fitting, with accompanying resources for teaching about modeling. As we suggested, a primary goal of our research effort is to make simulation-based models like our price accumulation approach accessible to researchers without strong statistical and computational backgrounds. This is outlined in Research Cluster #2, where the model creators (the PI &

research team) shoulder the computational burden of modeling by simulating a large volume of data from a model to examine how its parameters relate to observed data. Our lab has successfully implemented this approach now for several dynamic models of decision making and pricing, including Wald accumulator, diffusion, and basic price accumulation models [18,89,90] (see **Figure 2**). This proposal would support the creation of parameter estimation networks for any new modifications we make to these pricing models (see Research Plan). These networks will be embedded into online tools (using server-side Python applications) that allow researchers to upload their pricing data (e.g., in a CSV file) and instantaneously obtain individual-level parameter estimates. The network itself along with the online implementation will be made freely available on the Open Science.

Once developed, these tools will be incorporated into both graduate and undergraduate education. The PI has designed web-based interactive elements to accompany course content on different topics in decision-making [92] and incorporated these interactive elements into an undergraduate course on decision-making and a graduate course on cognitive modeling. The tools developed in Research Cluster #2 (Study 2) will be used as online interactive activities that provide (1) feedback about how users' (students') behavior compares to that of other participants, constantly updated as new data arrives from additional participants; (2) educational resources to visitors aimed at helping them understand what the experiment is measuring and how; and (3) a constant source of experimental results that can be used in research. Participants (students) will receive feedback about their risk / delay preferences based on the estimates of the price modeling network (**Figure 6**), accompanied by a simple explanation of what those parameters mean and how they are estimated. These resources have been shown to improve course engagement and students' understanding [93-95], and their incorporation into the course will be assessed through student feedback and reviews.

2. Integration of online interactive educational tools into financial education through the UF IFAS Extension Money Matters [MM] and First Time Homebuyers [FTH] classes. Once the online tools are fully developed, they will be deployed and integrated into not only undergraduate and graduate courses, but also outreach and extension learning programs. The University of Florida IFAS Extension holds online programs and in-person classes in all 67 Florida counties aimed at helping residents find employment and save for future events like college or retirement ("Money Matters" program) and provide financial education to first-time home buyers ("First Time Homebuyers" program). We will integrate the online risk / delay preference assessment tool into the curriculum of these classes, with the aim of assisting participants in understanding their own money management preferences and how these preferences align with their spending or saving goals. This will be combined with resources from the Money Matters [MM] program that helps participants create budgets and savings accounts (for example). This would allow us to personalize the MM program to individual participants; for example, a person who is highly risk-averse and has a shallow discounting rate might be comfortable developing safe, long-term investments like treasury bonds. Personalized tools for investment recommendations have long been used by banks to assess people's investment preferences [96] and create personalized financial plans [97]. Therefore, we expect that personalized plans or recommendations based on model parameters will have a positive impact on savings behavior for participants saving for down payments on houses [98].

This proposal would provide bidirectional benefits from integrating our research tools with the Money Matters program in several ways. First, its use in the local programs would allow us to track, over time, how participants in the Money Matters program think about risks and delays as they participate in financial education programs. As they gain financial literacy during the program, our pricing tools can be used to

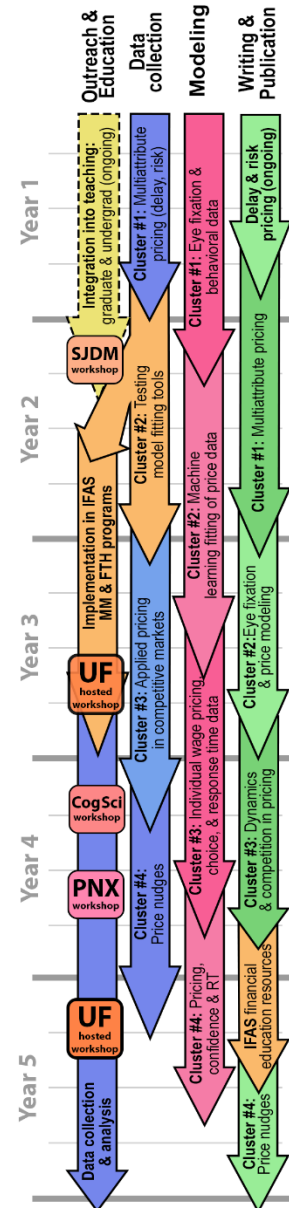


Figure 8: Outline of the timeline of research and education activities.

evaluate how their risk and delay preferences change over time, evaluating the impact it has on how they view money, risk, time, and their relation to saving. Research Cluster #3 provides empirical data that can help jobseekers and homebuyers maximize their chances of finding a job or a home, by evaluating the optimal level of strictness, importance of a budget, and relative importance of different job or house features in determining how satisfied and successful our participants were with their wage demands or home bids. It also suggests future avenues for interventions that can be incorporated into the financial education programs more formally. For example, excessive risk aversion or delay aversion among lower-income participants may signal the need for educational interventions that address financial anxiety [99], focus on developing different types of investment portfolios [100], or additional guidance on long-term saving [97].

Integration of these assessment tools into the IFAS website allows us to direct participants in online IFAS programs to the savings tools and resources that have already been developed and provided on the site. By providing an appealing introduction to money management through a “personality quiz” and

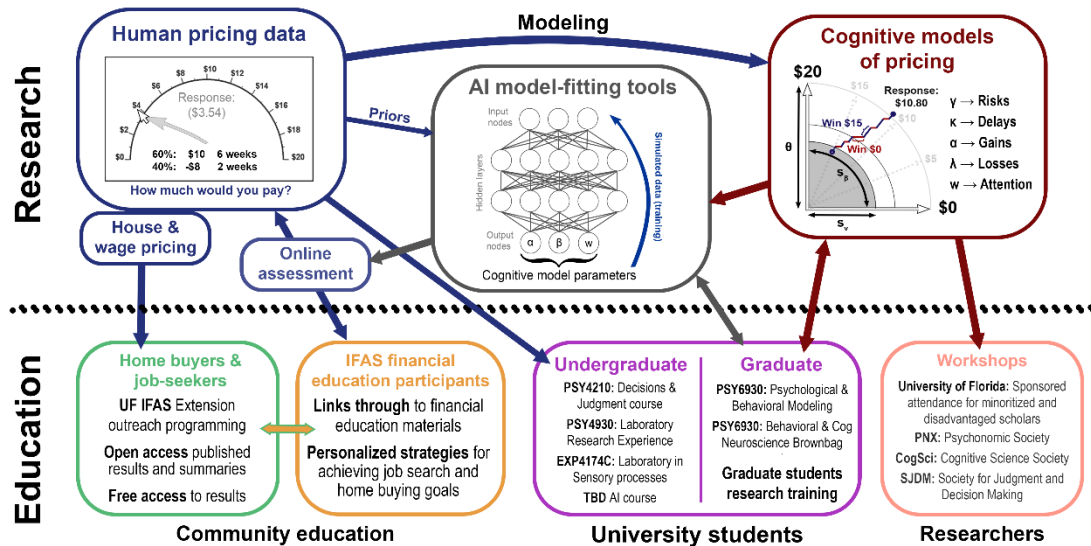


Figure 2: Diagram of the relationships and integration between research (top panels) and education & outreach activities (bottom panels) that are part of this proposal.

recommendation system, we can lower the barriers to entry on financial education and provide increased user engagement and enjoyment [101-103], thus improving the rate of participation and likelihood that online users continue through to the other IFAS money management materials. Success on this integration will be assessed by changes in traffic to Money Matters and First Time Homebuyers materials before vs after integration with the online assessment tools, and through structured feedback / interviews that are already conducted with participants in the programs. For example, participants are asked at the beginning and at the conclusion of an IFAS program about how likely they are to open a savings account, whether they feel better equipped to balance their finances, how likely they are to change their spending or savings habits, and whether they feel better informed about money management and their own preferences for savings and investment. This will be supplemented with the post-experiment surveys tracking participant success (Research Cluster #3, Study 2) following their participation in the financial education programs.

3. Local and conference-adjacent workshops on computational modeling and the use of amortized inference modeling tools. For researchers, the ability to immediately obtain best-fit posterior parameter estimates using our online tools should be an attractive prospect, and machine learning approaches to estimation are ones that can be implemented in many different areas. Therefore, an integral step in disseminating and enabling the use of modeling tools is to provide education on their structure, goals, and implementation. Recognizing that subjective value and pricing are interdisciplinary topics, we will reach a wide audience through conference workshops for audiences in psychology (Psychonomics / PNIX), cognitive science (Cognitive Science Society / CogSci), and decision sciences more generally (Society for Judgment and Decision Making / SJDM). In addition, in the third and fifth years of the project period, we will host a week-long local workshop at the University of Florida aimed at education in cognitive modeling

and machine learning among researchers who have not had extensive statistical or computational training. Special diversity fellowships for participating in this local workshop will be awarded to women and minoritized scholars who apply, to counter structural disadvantages these groups have faced in breaking into STEM fields. A workshop timeline is outlined in the left column of **Figure 8**.

The goal of these workshops will be tailored to the particular audience and the length of time allocated for each one, but it will focus on developing (1) a base of statistical and mathematical training, (2) understanding of how these relate to model fitting and comparison, (3) how simulation-based methods can be used to generate approximate likelihoods [104,105] and (4) how these methods can be used to train neural networks for model fitting like the ones we develop during the project period. To assess the effectiveness of these workshops, participants will be asked to complete pre-workshop, immediately post-workshop, and 1-year post-workshop surveys aimed at assessing their (self-reported) understanding of computational cognitive modeling, their anxiety about / ability to engage with research using models, how they use the tools or code we share and how they use modeling in their research more generally, and the degree to which they found the workshop(s) useful or educational. This will allow us to iteratively improve the content of the workshop and assess impact on the adoption of cognitive modeling in each field.

4. Public-facing educational materials on pricing and wages will be incorporated into IFAS guides and curricula. Our collaboration with UF's IFAS Extension program offers the opportunity to provide materials that are frequently used by local communities in Florida, including both homebuyers and jobseekers. As outlined above, the research we do in Clusters #2-3 of the Research Plan will be closely integrated with our outreach and education efforts related to wage and home prices. These results will be prepared in simple, accessible, public-facing reports that summarize our findings and provide corresponding advice (see, e.g., [63-66,106]). For homebuyers, this will help them making plans for saving for a down payment, understand the role of interest rates, and succeed in competitive housing markets. For jobseekers, the documents will detail the wages that their peers receive / desire in order to be fairly compensated for different aspects of their jobs. As a result, these documents will promote wage equity by creating transparency in the relationship between job characteristics and compensation.

Broader impacts

The research and education / outreach described above can be expected to have a significant impact on graduate and undergraduate education at UF, the research programs of participants in our workshops, improve financial education in IFAS Extension programming, and local jobseekers and first-time homebuyers. The materials that we develop and publish will enable jobseekers to better manage their wage request, help homebuyers make bids that are within their budget and likely to be accepted, and even help employers make changes to their employment structure that will attract jobseekers, such as building in additional safety measures, adjusting the job responsibilities, or changing the salary or hourly rate. Research has shown that this type of transparency and education is effective at reducing demographic wage gaps in the workplace [127-129], indicating that making these data public will help promote equitable wage structures and assist disadvantaged jobseekers in negotiating for a fair wage [130-133]. Furthermore, by integrating them into financial education programs that are already offered through UF IFAS Extensions, we can directly equip individuals and families with the tools they need to maintain healthy financial habits and achieve long-term savings goals (leveraging the nudges from Research Cluster #4).

The effects of the pandemic on employment and housing among different social groups has not been evenly distributed, making education on wages and home pricing an issue of equity. Older workers, Black and Asian workers, and unmarried mothers were the most likely to lose their jobs during lockdowns, and they have been the least likely to return to work in the past two years, even after controlling for education [3]. Understanding the individual differences in risks (safety concerns), family care, and other considerations that these groups face when searching for jobs is paramount to creating inclusive work environments that allow them to return to the labor force and employment. Finding low-cost ways to match employees with fulfilling positions by understanding the factors determining wage demands will help protect further businesses from closing, reducing the burden particularly on small, minority-owned businesses [1].

Black and Latino communities as well as veterans [142] have also been hit the hardest by increasing rents and home prices, resulting in a greater rate of evictions and lower rates of new home ownership [144]. This housing crisis has had detrimental effects on the physical and mental health of people living in these

affected communities [141]. As a result of high rent and home prices, low-income individuals are often forced into high-crime neighborhoods, food insecurity, issues with mental health, developmental disabilities, and a much higher risk of homelessness [145]. Increasing access to resources like First Time Homebuyers, and the accompanying “Renters rights and foreclosure” programs offered by IFAS, will help people move into affordable, permanent housing and avoid the risks associated with eviction, reducing the burden not only on these individuals but on social support systems in the United States more generally [144-147].

Our research results and outreach will also inform policy on minimum wages and unemployment benefits. Greater unemployment benefits naturally create greater reservation wages, serving as a reference point [107-109] for the marginal benefit of returning to work that cannot be fully reset when unemployment benefits are cut [110]. As a result, we can expect that reservation wages have been inexorably altered both by the expansion of unemployment as well as the health and safety concerns for essential workers that were introduced by the pandemic [111]. New minimum wage standards must therefore respond to not only the objective changes in job risks and responsibilities, but to the perceived value of labor (and relatively greater demand) from the jobseeker’s point of view [137]. Our pricing model – which disentangles reference points like reservation wages from job-specific wage factors – will quantify jobseekers’ greatest concerns related to their wages (Research Cluster #3), as well as protect jobseekers and homebuyers from exploitation with stronger labor regulations and consumer protection laws.

This project will also create tools that make computational modeling more accessible in psychology and cognitive and decision sciences. The amortized inference approach offloads the computational and statistical burden from the model user to the model creator (in this case, the PI and research team), meaning that even people without extensive statistical training can use our model fitting tools. Combined with our workshops aimed at teaching these methods to researchers, this approach will make cognitive modeling more ubiquitous and accessible, improving the science we generate as a field by making our most powerful modeling methods available to our most disadvantaged scholars. The fellowships we offer to attend these trainings will directly promote STEM training among women and minoritized researchers, promoting equity and diversity in the cognitive and decision sciences.

Summary

In terms of intellectual merits, proposed project will advance our understanding of pricing by creating empirical behavioral and eye tracking data about multi-attribute pricing problems, making models of pricing more accessible, applying this model to understanding home buying and job seeking, and identifying scenarios where pricing can improve people’s inferences and long-term savings outcomes. It will create new tools for researchers to use for fitting price models, create online interactive educational materials, and develop new materials for financial education programs. It synthesizes research and education goals by putting a cognitive model front and center (Figure 9), enabling unique new insights into the psychological antecedents of subjective value while promoting accessibility and beneficent applications of this research.

Previous NSF Support and Long-term Goals

A major goal of the PI’s research program is to understand the cognitive processes underlying selections among many alternatives and continuous judgments. The PI has previously received a Graduate Research Fellowship (DGE-1424871, approximately \$130,000) and supplementary GROW Australia award (\$20,000) from NSF from 06/2014 to 05/2017. This award supported graduate training and multiple projects leading to 8 journal publications [67,79,112-118], 5 book chapters / proceedings / technical reports [119-123], 3 workshops on computational modeling (2014-2017), and 19 invited talks and conference presentations. This proposed project would extend this effort to pricing, furthering our understanding how graded judgments of value are constructed and how they relate to real-world problems like wages. It would also capitalize on the PI’s experience giving workshops on computational modeling, simplifying concepts like quantum mechanics and probability theory for a research audience. This project would also tie the PI’s work into major NSF Big Ideas like the Future of Work at the Human-Technology Frontier (How do we pay people for labor in an ever-changing employment market? And how can we equip people with simple computational tools for handling complex data?) and Growing Convergence Research (combining work from psychology, cognitive science, economics, and business to create interdisciplinary research products).

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NSF BIOGRAPHICAL SKETCH

NAME: Kvam, Peter

POSITION TITLE & INSTITUTION: Assistant Professor, University of Florida

(a) PROFESSIONAL PREPARATION -(see PAPPG Chapter II.C.2.f.(a))

INSTITUTION	LOCATION	MAJOR / AREA OF STUDY	DEGREE (if applicable)	YEAR YYYY
Indiana University	Bloomington, IN	Mathematics, Sociology	BA	2012
Indiana University	Bloomington, IN	Psychology	BS	2012
Michigan State University	East Lansing, MI	Psychology	MA	2014
Michigan State University	East Lansing, MI	Psychology (Cognition & Cognitive Neuroscience, Concentration in Quantitative Methods)	PHD	2017
Max Planck Institute for Human Development University of Tasmania	Berlin, Berlin	Center for Adaptive Rationality	Graduate Student	2014 - 2015
	Hobart, TAS	Cognitive Modeling	Graduate Student	2017 - 2017
Indiana University	Bloomington, IN	Models of judgment and decision making	Postdoctoral Fellow	2017 - 2018
The Ohio State University	Columbus, OH	Model-based cognitive neuroscience	Postdoctoral Fellow	2018 - 2019

(b) APPOINTMENTS -(see PAPPG Chapter II.C.2.f.(b))

2019 - present Assistant Professor, University of Florida, Gainesville, FL

2018 - 2019 Postdoctoral Researcher, The Ohio State University, Columbus, OH

2017 - 2018 Research Associate, Indiana University, Bloomington, IN

(c) PRODUCTS -(see PAPPG Chapter II.C.2.f.(c))**Products Most Closely Related to the Proposed Project**

1. Kvam PD, Turner BM. Reconciling similarity across models of continuous selections. Psychol Rev. 2021 Jul;128(4):766-786. PubMed PMID: [34081510](https://pubmed.ncbi.nlm.nih.gov/34081510/).
2. Kvam P. A geometric framework for modeling dynamic decisions among arbitrarily many alternatives. Journal of Mathematical Psychology. 2019 August; 91:14-37. Available from: <https://linkinghub.elsevier.com/retrieve/pii/S0022249617302481> DOI: 10.1016/j.jmp.2019.03.001
3. Kvam P, Busemeyer J. A distributional and dynamic theory of pricing and preference. Psychological Review. 2020 November; 127(6):1053-1078. Available from: <http://doi.apa.org/getdoi.cfm?doi=10.1037/rev0000215> DOI: 10.1037/rev0000215
4. Kvam P, Romeu R, Turner B, Vassileva J, Busemeyer J. Testing the factor structure underlying

behavior using joint cognitive models: Impulsivity in delay discounting and Cambridge gambling tasks. *Psychological Methods*. 2021 February; 26(1):18-37. Available from: <http://doi.apa.org/getdoi.cfm?doi=10.1037/met0000264> DOI: 10.1037/met0000264

5. Reynolds A, Garton R, Kvam P, Sauer J, Osth A, Heathcote A. A dynamic model of deciding not to choose. *Journal of Experimental Psychology: General*. 2021 January; 150(1):42-66. Available from: <http://doi.apa.org/getdoi.cfm?doi=10.1037/xge0000770> DOI: 10.1037/xge0000770

Other Significant Products, Whether or Not Related to the Proposed Project

1. Kvam PD, Pleskac TJ. Strength and weight: The determinants of choice and confidence. *Cognition*. 2016 Jul;152:170-180. PubMed PMID: [27093221](#).
2. Kvam PD. Modeling accuracy, response time, and bias in continuous orientation judgments. *J Exp Psychol Hum Percept Perform*. 2019 Mar;45(3):301-318. PubMed PMID: [30714760](#).
3. Kvam PD, Bussemeyer JR, Pleskac TJ. Temporal oscillations in preference strength provide evidence for an open system model of constructed preference. *Sci Rep*. 2021 Apr 14;11(1):8169. PubMed Central PMCID: [PMC8046775](#).
4. Kvam P, Pleskac T, Yu S, Bussemeyer J. Interference effects of choice on confidence: Quantum characteristics of evidence accumulation. *Proceedings of the National Academy of Sciences*. 2015 August 10; 112(34):10645-10650. Available from: <https://pnas.org/doi/full/10.1073/pnas.1500688112> DOI: 10.1073/pnas.1500688112
5. Kvam P, Pleskac T. Strength and weight: The determinants of choice and confidence. *Cognition*. 2016 July; 152:170-180. Available from: <https://linkinghub.elsevier.com/retrieve/pii/S0010027716300968> DOI: 10.1016/j.cognition.2016.04.008

(d) SYNERGISTIC ACTIVITIES -(see PAPPG Chapter II.C.2.f.(d))

1. Developed and taught a new undergraduate-level course, "Decisions and Judgment" at the University of Florida (offered as special topics). This included creating course assignments, quizzes, exams, and publicly-available interactive materials to accompany readings.
2. Designed and taught a new graduate-level course "Psychological and Behavioral Modeling" in the Department of Psychology at the University of Florida. This included creating course materials, assignments, exams, programming / coding activities, and supervising student research projects implementing computational modeling.
3. Served as a faculty mentor in the Undergraduate Multicultural Mentoring Program at the University of Florida, which provides one-on-one faculty mentoring for first-generation and minority college students, training and career development resources, and opportunities for involvement in research in the faculty member's laboratory.
4. Member of the Education Committee for Center for Behavioral Economics in Health Research at the University of Florida, which organizes graduate and professional training events, evaluates funding proposals for graduate dissertation research, and hosts workshops on new methods in Behavioral Economics that may be useful for researchers across various areas of health research.
5. Organizer and presenter for Full day tutorial on Quantum Models of Cognition and Decision,

The 36th/37th/38th/40th Annual Meetings of the Cognitive Science Society.

Other Personnel Biographical Information

Data Not Available

SUMMARY PROPOSAL BUDGET

YEAR 1

ORGANIZATION University of Florida				FOR NSF USE ONLY			
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Peter Kvam				PROPOSAL NO. 2237119	DURATION (months)		
				AWARD NO.	Proposed	Granted	
A. SENIOR PERSONNEL: PI/PI, Co-PI's, Faculty and Other Senior Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer	Funds granted by NSF (if different)
				CAL	ACAD	SUMR	
1. Peter Kvam - Principal Inv				1.0			9,466
2.							
3.							
4.							
5.							
6. () OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)				0.0			0
7. (1) TOTAL SENIOR PERSONNEL (1 - 6)				1.0			9,466
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)							
1. (0) POST DOCTORAL SCHOLARS				0.0			0
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				0.0			0
3. (1) GRADUATE STUDENTS							23,221
4. (0) UNDERGRADUATE STUDENTS							0
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)							0
6. (4) OTHER							5,058
TOTAL SALARIES AND WAGES (A + B)							37,745
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)							7,380
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)							45,125
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$5,000.) HiPerGator				\$ 0.0			
TOTAL EQUIPMENT							0
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)							7,560
2. INTERNATIONAL							0
F. PARTICIPANT SUPPORT COSTS							
1. STIPENDS \$ 0							
2. TRAVEL 0							
3. SUBSISTENCE 0							
4. OTHER 0							
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS							0
G. OTHER DIRECT COSTS							
1. MATERIALS AND SUPPLIES							3,161
2. PUBLICATION COSTS/DOCUMENTATION/DISSEMINATION							0
3. CONSULTANT SERVICES							0
4. COMPUTER SERVICES							0
5. SUBAWARDS							0
6. OTHER							23,815
TOTAL OTHER DIRECT COSTS							26,976
H. TOTAL DIRECT COSTS (A THROUGH G)							79,661
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE) MTDC (Rate: 52.5, Base:66946.0)							
TOTAL INDIRECT COSTS (F&A)							35,147
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)							114,808
K. FEE							0
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)							114,808
M. COST SHARING PROPOSED LEVEL \$ 0				AGREED LEVEL IF DIFFERENT \$			
PI/PI NAME Peter Kvam				FOR NSF USE ONLY			
ORG. REP. NAME* Angela Cocchiarella				INDIRECT COST RATE VERIFICATION			
				Date Checked	Date Of Rate Sheet	Initials - ORG	

1 *ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

SUMMARY PROPOSAL BUDGET

YEAR 2

ORGANIZATION University of Florida				FOR NSF USE ONLY			
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Peter Kvam				PROPOSAL NO. 2237119	DURATION (months)		
				AWARD NO.	Proposed	Granted	
A. SENIOR PERSONNEL: PI/PI, Co-PI's, Faculty and Other Senior Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer	Funds granted by NSF (if different)
				CAL	ACAD	SUMR	
1. Peter Kvam - Principal Inv				1.5			14,626
2.							
3.							
4.							
5.							
6. () OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)				0.0			0
7. (1) TOTAL SENIOR PERSONNEL (1 - 6)				1.5			14,626
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)							
1. (0) POST DOCTORAL SCHOLARS				0.0			0
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				0.0			0
3. (1) GRADUATE STUDENTS							23,918
4. (0) UNDERGRADUATE STUDENTS							0
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)							0
6. (4) OTHER							5,211
TOTAL SALARIES AND WAGES (A + B)							43,755
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)							9,116
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)							52,871
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$5,000.) HiPerGator							
TOTAL EQUIPMENT							0
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)							7,560
2. INTERNATIONAL							0
F. PARTICIPANT SUPPORT COSTS							
1. STIPENDS \$ 0							0
2. TRAVEL 0							0
3. SUBSISTENCE 0							0
4. OTHER 0							0
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS							0
G. OTHER DIRECT COSTS							
1. MATERIALS AND SUPPLIES							0
2. PUBLICATION COSTS/DOCUMENTATION/DISEMINATION							0
3. CONSULTANT SERVICES							0
4. COMPUTER SERVICES							0
5. SUBAWARDS							0
6. OTHER							55,498
TOTAL OTHER DIRECT COSTS							55,498
H. TOTAL DIRECT COSTS (A THROUGH G)							115,929
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE) MTDC (Rate: 52.5, Base:102431.0)							
TOTAL INDIRECT COSTS (F&A)							53,776
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)							169,705
K. FEE							0
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)							169,705
M. COST SHARING PROPOSED LEVEL \$ 0				AGREED LEVEL IF DIFFERENT \$			
PI/PI NAME Peter Kvam				FOR NSF USE ONLY			
ORG. REP. NAME* Angela Cocchiarella				INDIRECT COST RATE VERIFICATION			
				Date Checked	Date Of Rate Sheet	Initials - ORG	

2 *ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

SUMMARY PROPOSAL BUDGET

YEAR 3

ORGANIZATION University of Florida				FOR NSF USE ONLY			
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Peter Kvam				PROPOSAL NO. 2237119	DURATION (months)		
				AWARD NO.	Proposed	Granted	
A. SENIOR PERSONNEL: PI/PI, Co-PI's, Faculty and Other Senior Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer	Funds granted by NSF (if different)
				CAL	ACAD	SUMR	
1. Peter Kvam - Principal Inv				1.5			15,065
2.							
3.							
4.							
5.							
6. () OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)				0.0			0
7. (1) TOTAL SENIOR PERSONNEL (1 - 6)				1.5			15,065
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)							
1. (0) POST DOCTORAL SCHOLARS				0.0			0
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				0.0			0
3. (1) GRADUATE STUDENTS							24,636
4. (0) UNDERGRADUATE STUDENTS							0
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)							0
6. (4) OTHER							5,366
TOTAL SALARIES AND WAGES (A + B)							45,067
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)							9,389
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)							54,456
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$5,000.) HiPerGator							\$ 3600.0
TOTAL EQUIPMENT							3,600
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)							7,560
2. INTERNATIONAL							0
F. PARTICIPANT SUPPORT COSTS							
1. STIPENDS \$ 4,000							0
2. TRAVEL							0
3. SUBSISTENCE							0
4. OTHER							0
TOTAL NUMBER OF PARTICIPANTS (5) TOTAL PARTICIPANT COSTS							4,000
G. OTHER DIRECT COSTS							
1. MATERIALS AND SUPPLIES							4,000
2. PUBLICATION COSTS/DOCUMENTATION/DISEMINATION							0
3. CONSULTANT SERVICES							0
4. COMPUTER SERVICES							0
5. SUBAWARDS							0
6. OTHER							56,429
TOTAL OTHER DIRECT COSTS							60,429
H. TOTAL DIRECT COSTS (A THROUGH G)							130,045
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE) MTDC (Rate: 52.5, Base:108272.0)							
TOTAL INDIRECT COSTS (F&A)							56,843
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)							186,888
K. FEE							0
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)							186,888
M. COST SHARING PROPOSED LEVEL \$ 0				AGREED LEVEL IF DIFFERENT \$			
PI/PI NAME Peter Kvam				FOR NSF USE ONLY			
ORG. REP. NAME* Angela Cocchiarella				INDIRECT COST RATE VERIFICATION			
				Date Checked	Date Of Rate Sheet	Initials - ORG	

3 *ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

SUMMARY PROPOSAL BUDGET

YEAR 4

ORGANIZATION University of Florida				FOR NSF USE ONLY			
				PROPOSAL NO. 2237119	DURATION (months)		
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Peter Kvam				AWARD NO.	Proposed	Granted	
A. SENIOR PERSONNEL: PI/PI, Co-PI's, Faculty and Other Senior Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer	Funds granted by NSF (if different)
				CAL	ACAD	SUMR	
1. Peter Kvam - Principal Inv				1.5			15,516
2.							
3.							
4.							
5.							
6. () OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)				0.0			0
7. (1) TOTAL SENIOR PERSONNEL (1 - 6)				1.5			15,516
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)							
1. (0) POST DOCTORAL SCHOLARS				0.0			0
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				0.0			0
3. (1) GRADUATE STUDENTS							25,375
4. (0) UNDERGRADUATE STUDENTS							0
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)							0
6. (4) OTHER							5,527
TOTAL SALARIES AND WAGES (A + B)							46,418
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)							9,670
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)							56,088
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$5,000.) HiPerGator							\$ 0.0
TOTAL EQUIPMENT							0
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)							7,560
2. INTERNATIONAL							0
F. PARTICIPANT SUPPORT COSTS							
1. STIPENDS \$ 0							0
2. TRAVEL 0							0
3. SUBSISTENCE 0							0
4. OTHER 0							0
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS							0
G. OTHER DIRECT COSTS							
1. MATERIALS AND SUPPLIES							0
2. PUBLICATION COSTS/DOCUMENTATION/DISEMINATION							0
3. CONSULTANT SERVICES							0
4. COMPUTER SERVICES							0
5. SUBAWARDS							0
6. OTHER							18,281
TOTAL OTHER DIRECT COSTS							18,281
H. TOTAL DIRECT COSTS (A THROUGH G)							81,929
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE) MTDC (Rate: 52.5, Base:67048.0)							
TOTAL INDIRECT COSTS (F&A)							35,200
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)							117,129
K. FEE							0
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)							117,129
M. COST SHARING PROPOSED LEVEL \$ 0				AGREED LEVEL IF DIFFERENT \$			
PI/PI NAME Peter Kvam				FOR NSF USE ONLY			
ORG. REP. NAME* Angela Cocchiarella				INDIRECT COST RATE VERIFICATION			
				Date Checked	Date Of Rate Sheet	Initials - ORG	

4 *ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

SUMMARY PROPOSAL BUDGET

YEAR 5

ORGANIZATION University of Florida				FOR NSF USE ONLY			
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Peter Kvam				PROPOSAL NO. 2237119	DURATION (months)		
				AWARD NO.	Proposed	Granted	
A. SENIOR PERSONNEL: PI/PI, Co-PI's, Faculty and Other Senior Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer	Funds granted by NSF (if different)
				CAL	ACAD	SUMR	
1. Peter Kvam - Principal Inv				1.5			15,982
2.							
3.							
4.							
5.							
6. () OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)				0.0			0
7. (1) TOTAL SENIOR PERSONNEL (1 - 6)				1.5			15,982
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)							
1. (0) POST DOCTORAL SCHOLARS				0.0			0
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				0.0			0
3. (1) GRADUATE STUDENTS							26,136
4. (0) UNDERGRADUATE STUDENTS							0
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)							0
6. (0) OTHER							0
TOTAL SALARIES AND WAGES (A + B)							42,118
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)							8,195
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)							50,313
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$5,000.) HiPerGator							\$ 0.0
TOTAL EQUIPMENT							0
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)							7,560
2. INTERNATIONAL							0
F. PARTICIPANT SUPPORT COSTS							
1. STIPENDS \$ 4,000							0
2. TRAVEL							0
3. SUBSISTENCE							0
4. OTHER							0
TOTAL NUMBER OF PARTICIPANTS (5) TOTAL PARTICIPANT COSTS							4,000
G. OTHER DIRECT COSTS							
1. MATERIALS AND SUPPLIES							0
2. PUBLICATION COSTS/DOCUMENTATION/DISEMINATION							0
3. CONSULTANT SERVICES							0
4. COMPUTER SERVICES							0
5. SUBAWARDS							0
6. OTHER							15,881
TOTAL OTHER DIRECT COSTS							15,881
H. TOTAL DIRECT COSTS (A THROUGH G)							77,754
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE) MTDC (Rate: 52.5, Base:58129.0)							
TOTAL INDIRECT COSTS (F&A)							30,518
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)							108,272
K. FEE							0
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)							108,272
M. COST SHARING PROPOSED LEVEL \$ 0				AGREED LEVEL IF DIFFERENT \$			
PI/PI NAME Peter Kvam				FOR NSF USE ONLY			
ORG. REP. NAME* Angela Cocchiarella				INDIRECT COST RATE VERIFICATION			
				Date Checked	Date Of Rate Sheet	Initials - ORG	

5 *ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

SUMMARY PROPOSAL BUDGET

Cumulative

ORGANIZATION University of Florida				FOR NSF USE ONLY			
				PROPOSAL NO. 2237119	DURATION (months)		
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Peter Kvam				AWARD NO.	Proposed	Granted	
A. SENIOR PERSONNEL: PI/PI, Co-PI's, Faculty and Other Senior Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer	Funds granted by NSF (if different)
				CAL	ACAD	SUMR	
1. Peter Kvam - Principal Inv				7.0			70,655
2.							
3.							
4.							
5.							
6. () OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)							
7. (1) TOTAL SENIOR PERSONNEL (1 - 6)				7.0			70,655
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)							
1. (0) POST DOCTORAL SCHOLARS				0.0			0
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				0.0			0
3. (5) GRADUATE STUDENTS							123,286
4. (0) UNDERGRADUATE STUDENTS							0
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)							0
6. (16) OTHER							21,162
TOTAL SALARIES AND WAGES (A + B)							215,103
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)							43,750
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)							258,853
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$5,000.)							
TOTAL EQUIPMENT							3,600
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)							37,800
2. INTERNATIONAL							0
F. PARTICIPANT SUPPORT COSTS							
1. STIPENDS \$ 8,000							
2. TRAVEL 0							
3. SUBSISTENCE 0							
4. OTHER 0							
TOTAL NUMBER OF PARTICIPANTS (10) TOTAL PARTICIPANT COSTS							8,000
G. OTHER DIRECT COSTS							
1. MATERIALS AND SUPPLIES							7,161
2. PUBLICATION COSTS/DOCUMENTATION/DISEMINATION							0
3. CONSULTANT SERVICES							0
4. COMPUTER SERVICES							0
5. SUBAWARDS							0
6. OTHER							169,904
TOTAL OTHER DIRECT COSTS							177,065
H. TOTAL DIRECT COSTS (A THROUGH G)							485,318
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE)							
TOTAL INDIRECT COSTS (F&A)							211,484
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)							696,802
K. FEE							0
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)							696,802
M. COST SHARING PROPOSED LEVEL \$ 0				AGREED LEVEL IF DIFFERENT \$			
PI/PI NAME Peter Kvam				FOR NSF USE ONLY			
ORG. REP. NAME* Angela Cocchiarella				INDIRECT COST RATE VERIFICATION			
				Date Checked	Date Of Rate Sheet	Initials - ORG	

C *ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

BUDGET JUSTIFICATION**SALARIES:**

The budget provides the anticipated effort for persons to be involved in this research project, if funded, for the total requested time period. For effort reporting, UF defines year as a calendar, 12-month year. For projected time occurring in future years, the salaries have been adjusted with a 3.0% increase for cost-of-living adjustments.

A. SENIOR PERSONNEL:

Peter Kvam, Ph.D., Principal Investigator (effort = YR 1 CM 1, YR 2-5 1.5 CM)): PI-Kvam will be involved in all aspects of the of the project, including planning and execution of experiments, analysis and modeling, organizing workshops, implementing and testing machine learning models, presenting, and reporting. His background and training in computational modeling and its application to studying the cognitive processes underlying decision making, including models that make inferences about underlying associations based on patterns of response times and accuracy, make him well-suited to designing, implementing, and providing guidance on the models proposed in the project. His expertise in computational modeling, online experiments, and machine learning will allow him to implement and train the PDR on implementation of the online model fitting and assessment tools. In Years 1-5, he will train the Graduate Research Assistant (GRA), supervising their involvement on the project as they carry out experiment design, data collection, modeling, and collaborate with IFAS personnel listed below. Throughout the course of the grant, PI-Kvam will share responsibility for analyzing data with the GRA – allowing the GRA to take on more responsibility as they develop statistical capabilities. PI-Kvam will also mentor the GRA through the writing process; the expectation is that the GRA will lead at least two manuscripts with the PI as co-author. PI-Kvam additionally expects to lead at least four additional papers based on these data. PI-Kvam and the GRA will present at national conferences on this data. PI-Kvam will also manage grant funds and ensure that the timeline is being followed, and be responsible for reporting to NSF.

If all current proposals are funded at the requested level, the PI's total salary amount across all of his funded projects will exceed the two months normally allowed by NSF for senior project personnel. However, the level of effort requested on this project and all of the PI's other funded projects represents the absolute minimum amount needed to complete the proposed scope of work for these projects. Any reduction in his effort to comply with the two-month rule would negatively impact his ability to complete the proposed work on time.

B. OTHER PERSONNEL:**Lisa Hamilton (effort = 0.5 CM / Year, Years 1-4)**

Lisa Hamilton is the University of Florida Institute for Food and Agricultural Sciences (IFAS) Extension Agent for Volusia County, Florida. She is in charge of running the financial education programs in her area. She will be responsible for recruiting participants from these financial education programs, answering any questions that they have, contacting participants as needed after they participate in the program (e.g., during Research Cluster 3-2), and integrating our research findings into ongoing financial education programs.

Johanna Gomez-Gonzalez (effort = 0.5 CM / Year, Years 1-4)

Johanna Gomez-Gonzalez is the University of Florida Institute for Food and Agricultural Sciences (IFAS) Extension Agent for Osceola County, Florida. She is in charge of delivering the financial education programs in her area. She will be responsible for recruiting participants from these financial education programs, answering any questions that they have, contacting

participants as needed after they participate in the program (e.g., during Research Cluster 3-2), and integrating our research findings into ongoing financial education programs. She will also assist with translating the experimental and survey materials as well as the research findings into Spanish for primarily Spanish-speaking participants in the program.

Jenny Rodriguez (effort = 0.5 CM / Year, Years 1-4)

Jenny Rodriguez is the University of Florida Institute for Food and Agricultural Sciences (IFAS) Extension Agent for Orange County, Florida. She is in charge of delivering the financial education programs in her area. She will be responsible for recruiting participants from these financial education programs, answering any questions that they have, contacting participants as needed after they participate in the program (e.g., during Research Cluster 3-2), and integrating our research findings into ongoing financial education programs. She will also assist with translating the experimental and survey materials as well as the research findings into Spanish for primarily Spanish-speaking participants in the program.

Katherine Marin (effort = 0.5 CM / Year, Years 1-4)

Katherine Marin is the University of Florida Institute for Food and Agricultural Sciences (IFAS) Extension Agent for Miami-Dade County, Florida. She is in charge of delivering the financial education programs in her area. She will be responsible for recruiting participants from these financial education programs, answering any questions that they have, contacting participants as needed after they participate in the program (e.g., during Research Cluster 3-2), and integrating our research findings into ongoing financial education programs. She will also assist with translating the experimental and survey materials as well as the research findings into Spanish for primarily Spanish-speaking participants in the program.

TBD, Graduate Student, (effort = 4.5 CM / Years 1-5)

Funds are requested to support one senior GRA at 4.5 calendar months. The GRA will assist the PI in creating experimental materials, programming experiments, analyzing data, and writing manuscripts. The GRA will supervise laboratory data collection by undergraduate research assistants. The GRA will be expected to present data from the proposed experiments at a conference in Years 1-3 (e.g., regional and/or national) and take the lead on writing at least two manuscripts for publication.

Please note at the University of Florida, Graduate Students are appointed at a 0.50 FTE but devote 100% of their project appointed effort toward a sponsored project(s) to complete the research credit requirements of their academic program.

C. FRINGE BENEFITS:

Fringe benefits, including FICA, State Unemployment, Workers' Compensation, Retirement, Life and Health Insurance, are assessed as a percentage of the respective employee's salary. Fringe benefits are calculated on the requested salary budgeted and charged in accordance with the University of Florida's negotiated and approved rate agreement; which can be found here: <https://research.ufl.edu/wp-content/uploads/FA-agreement.pdf>

D. EQUIPMENT:

UF Research Computing operates a cluster for research use by faculty and their research associates. The details of the operation are described under the Facilities section of this proposal. This includes a description of the resources made available by the University to support HPC-related research. By buying a part of the shared cluster at \$200 per NCU

(normalized computing unit) and \$1200 per NGU (NVIDIA GPU, which gives access to one NVIDIA RTX 2080ti GPU or Ampere A-100 computing time), our research project leverages the whole infrastructure, which includes system acquisition, configuration, installation, administration, maintenance, and application support by professional staff to the researchers working in this project. The purchase of NCUs (Normalized Compute Units), i.e. compute cores, and GPUs are viewed as an acquisition of hardware. Therefore, the University of Florida considers these costs to be capitalized equipment and as a result, the grant or contract is not charged IDC (indirect cost), also known as overhead.

Funds are requested in Year 3 to replace supercomputing resources (12 CPU cores, 1 GPU) in the lab that were purchased at the beginning of 2021 and will be reaching the end of their hardware lifecycle in Year 3 (2025-2026). These are used for parallel computing used in model fitting (pricing models) and simulation, as well as training the deep neural network that is used to approximate parameter estimates in the model. This hardware is maintained by UF Research Computing during its hardware lifetime. The price of this equipment is based on UF's HiPerGator hardware pricing: Total \$3,600.

<https://www.rc.ufl.edu/services/rates/hardware/>

E. TRAVEL:

All travel costs are consistent with the University of Florida's travel policy (located at <http://www.fa.ufl.edu/directives-and-procedures/travel/>).

Domestic:

Year	Conference	Location	Personnel support (days/nights)	Airfare & Ground Transport	Registration	Meals / Lodging	Total
1	MathPsych	TBA	PI (5/4)	\$750	\$150	\$980	\$1880
1	MathPsych	TBA	GRA (5/4)	\$750	\$50	\$980	\$1780
1	SJDM	TBA	PI (5/4)	\$750	\$350	\$980	\$2080
1	SJDM	TBA	GRA (5/4)	\$750	\$90	\$980	\$1820
2	MathPsych	TBA	PI (5/4)	\$750	\$150	\$980	\$1880
2	MathPsych	TBA	GRA (5/4)	\$750	\$50	\$980	\$1780
2	SJDM	TBA	PI (5/4)	\$750	\$350	\$980	\$2080
2	SJDM	TBA	GRA (5/4)	\$750	\$90	\$980	\$1820
3	MathPsych	TBA	PI (5/4)	\$750	\$150	\$980	\$1880
3	MathPsych	TBA	GRA (5/4)	\$750	\$50	\$980	\$1780
3	SJDM	TBA	PI (5/4)	\$750	\$350	\$980	\$2080
3	SJDM	TBA	GRA (5/4)	\$750	\$90	\$980	\$1820
4	CogSci	TBA	PI (5/4)	\$750	\$150	\$980	\$1880
4	CogSci	TBA	GRA (5/4)	\$750	\$50	\$980	\$1780
4	PNX	TBA	PI (5/4)	\$750	\$350	\$980	\$2080
4	PNX	TBA	GRA (5/4)	\$750	\$90	\$980	\$1820
5	MathPsych	TBA	PI (5/4)	\$750	\$150	\$980	\$1880
5	MathPsych	TBA	GRA (5/4)	\$750	\$50	\$980	\$1780
5	SJDM	TBA	PI (5/4)	\$750	\$350	\$980	\$2080
5	SJDM	TBA	GRA (5/4)	\$750	\$90	\$980	\$1820

Registration for Society for Judgment and Decision Making [SJDM] and Psychonomics [PNX]

annual conferences is estimated at slightly higher rates than the 2019 SJDM convention: \$500 for faculty members; \$250 for GRA. Registration for Mathematical Psychology [MathPsych] and Cognitive Science Society [CogSci] conferences are estimated at a (discounted presenter, for CogSci) rate of \$100 for faculty and \$50 for students.

Lodging at the conference hotel is estimated at \$200/night. Meals are at the per diem daily rate of \$36 / day for each attendee is as specified by the University of Florida:

<https://www.fa.ufl.edu/directives/per-diem/>

F. PARTICIPANT SUPPORT COSTS:

Travel – In Years 3 and 5, funds are requested to lead an onsite training workshop. Requested funds cover travel support for 5 workshop participants (average \$800 per participant) to support attendance. Priority travel support will be given to participants from minoritized groups and women, to counter structural inequalities that have put these groups at a disadvantage when pursuing advanced training in computational / STEM fields. Total \$4,000 each in Years 3 and 5.

G. OTHER DIRECT COSTS:

Computers

Funds are requested to purchase a computer sufficient for high-throughput modeling applications (MATLAB / Python, machine learning) in Year 1 of the proposal, and to replace two such modeling computers in the laboratory (purchased in 2019) that will be reaching the end of their hardware lifecycle in Year 3 (end of 5 years; 2025). This includes \$3,161 in Year 1 for a computer for the graduate RA, and \$4,000 in Year 3 to replace the PI's computer when it hits the end of its hardware lifecycle

Other – Reitz rental fees for workshops

The local workshop will be hosted in Years 3 and 5 at the University of Florida Reitz Union through the Office of Event Services. Funds are requested for room rental fees (\$38 / 1 room / day), A/V Equipment (LCD Projector, podium with microphone, screen; \$0), and a fee for set-up- (\$26 / day) for a total of \$256 for the full 4-day workshop. These follow the rental rates for the Reitz Union set out at <https://eventservices.ufsa.ufl.edu/our-spaces/rental-rates/>.

Other – Tuition: Tuition expenses will be charged in direct proportion to the total percentage of effort expended on this project. UF anticipates 10% increases on tuition until Fall 2024, when annual increases will decrease to 5% annually.

Other – Human Subject Payments:

Human subjects will be recruited to complete the studies in Research Clusters 1-4. As we state in the Project Description, this will involve recruiting human subjects as follows:

Year	Study number	Number of participants	Sessions per participant	Compensation	Total
1	1-1	85	1	~\$20/session	\$3400
1	1-3	85	1	~\$20/session	\$1700
2	2-1	85	1	\$0 (research credits)	\$0
2	2-2	Unlimited	1	\$0 (IFAS / online)	\$0
2-3	3-1	600	1	~\$20/session (IFAS / online)	\$12,000
2-3	3-2	600	24	~\$5/session	\$72,000

4	4-1	85	1	~\$20/hour	\$1700
4	4-2	85	1	~\$20/hour	\$1700

Other – Human Subjects recruitment service – Prolific Academic

Funds are requested in Year 1 to support recruitment of a sample of U.S. 300 residents through Prolific Academic. A quote is attached to this proposal outlining the cost of this proposal for participants who are paid at a rate of \$15 for 60 minutes of participation each. With Prolific's fees, the total cost for Study 1-2 is \$6000.

H. TOTAL DIRECT COSTS: \$485,318

I. INDIRECT COSTS: \$211,484

F&A rates are negotiated and determined by the Department of Health and Human Services (DHHS). The University currently has an approved F&A rate of 52.50% of Modified Total Direct Costs (MTDC) for on-campus organized research. Modified total direct costs, consisting of all direct salaries and wages, applicable fringe benefits, materials and supplies, services, travel, and up to the first \$25,000 of each subaward (regardless of the period of performance of the subawards under the award). Modified total direct costs shall exclude equipment, capital expenditures, charges for patient care, rental costs, tuition remission, scholarships and fellowships, participant support costs and the portion of each subaward in excess of \$25,000. The point of contact is Lucy Siow, (301) 492-4855 on UF's approved rate agreement, which can be found here: <https://research.ufl.edu/wp-content/uploads/FA-agreement.pdf>.

J. TOTAL DIRECT AND INDIRECT COSTS: \$696,802

Effective 10/04/2021 NSF CURRENT AND PENDING SUPPORT

OMB-3145-0058

PI/co-PI/Senior Personnel: Kvam, Peter

PROJECT/PROPOSAL CURRENT SUPPORT

1. Project/Proposal Title: Varieties of impulsivity in opiate and stimulant users

Proposal/Award Number (if available): 2R01DA021421-11A1

Source of Support: National Institutes of Health / NIDA

Primary Place of Performance: University of Florida

Project/Proposal Support Start Date (if available): 01/2022

Project/Proposal Support End Date (if available): 05/2026

Total Award Amount (including Indirect Costs): \$93,582

Person-Month(s) (or Partial Person-Months) Per Year Committed to the Project:

Year	Person-months per year committed
2022	0.96
2023	0.96
2024	0.96
2025	0.96
2026	0.96

Overall Objectives: To use computational cognitive modeling to quantify behavior, self-report, and neuroimaging data from individuals with substance use problems in terms of meaningful model parameters related to processes like motivation, impulsivity, cognitive control, memory, and sensitivity to risks, gains, and losses. The goal is to better understand the cognitive processes involved in specific types of substance use using data collected under the parent award R01DA021421.

Statement of Potential Overlap: None with present project

PROJECT/PROPOSAL PENDING SUPPORT

1. Project/Proposal Title: Theme 4: Decision Intelligence for AI-Augmented Actions to Outcomes Decision Improvement (DECISIVE)

Proposal/Award Number (if available):

Source of Support: National Science Foundation

Primary Place of Performance: University of Florida

Project/Proposal Support Start Date (if available): 06/2023

Project/Proposal Support End Date (if available): 05/2028

Total Award Amount (including Indirect Costs): \$380,844

Person-Month(s) (or Partial Person-Months) Per Year Committed to the Project:

Year	Person-months per year committed
2024	0.67
2025	0.67
2026	0.67
2027	0.67
2028	0.67

Overall Objectives: The objective of the Decision Intelligence for AI-Augmented Actions to Outcomes Decision Improvement (DECISIVE) Institute is to advance foundational AI principles, techniques, and tools for democratized AI-augmented action-to-outcome decision making. Specifically, the component at UF is aimed at understanding basic decision processes and how AI can be used to evaluate and emulate human behavior by allowing it to simulate cognitive models.

Statement of Potential Overlap: None with the current project.

2. Project/Proposal Title: CAREER: An accessible, dynamic model of pricing applied to wage and housing purchases (this proposal)

Proposal/Award Number (if available):

Source of Support: National Science Foundation

Primary Place of Performance: University of Florida

Project/Proposal Support Start Date (if available): 05/2023

Project/Proposal Support End Date (if available): 04/2028

Total Award Amount (including Indirect Costs): \$696,649

Person-Month(s) (or Partial Person-Months) Per Year Committed to the Project:

Year	Person-months per year committed
2024	1
2025	1.5
2026	1.5
2027	1.5
2028	1.5

Overall Objectives: The goal of this proposal is to create a program of research on pricing, addressing both active research questions on subjective value and integrating research results into financial education programs, graduate and undergraduate education, and continuing education programs for active researchers in judgement and decision making. It will create accessible tools for computational modeling and apply these to understanding how people assign value in housing and employment markets.

Statement of Potential Overlap: None - this proposal.

3. Project/Proposal Title: Equipping AI with cognitive models for intent inference

Proposal/Award Number (if available):

Source of Support: Air Force Office of Scientific Research

Primary Place of Performance: University of Florida

Project/Proposal Support Start Date (if available): 01/2023

Project/Proposal Support End Date (if available): 01/2026

Total Award Amount (including Indirect Costs): \$416,003

Person-Month(s) (or Partial Person-Months) Per Year Committed to the Project:

Year	Person-months per year committed
2023	1
2024	1
2025	1.5

Overall Objectives: To create AI that can make inferences about latent motives by comparing observed behavior to cognitive models of motivation, and to test interventions where this AI is deployed to assist human decision-makers

Statement of Potential Overlap: None with current proposal

4. Project/Proposal Title: Equipping AI with cognitive models for decision support and intent inference

Proposal/Award Number (if available):

Source of Support: Office of Naval Research

Primary Place of Performance: University of Florida

Project/Proposal Support Start Date (if available): 10/2022

Project/Proposal Support End Date (if available): 09/2025

Total Award Amount (including Indirect Costs): \$712,538

Person-Month(s) (or Partial Person-Months) Per Year Committed to the Project:

Year	Person-months per year committed
2023	1.5
2024	2
2025	2

Overall Objectives: The goal of this proposal is to create and test systems that can infer a person's intent from their observed behavior. It is aimed at creating AI that can determine which of a set of goals a person is pursuing, use that information to support a human decision maker, and to embed this AI within autonomous agents to allow them to make better decisions on a continuous-time, interactive task.

Statement of Potential Overlap: None with current proposal

5. Project/Proposal Title: Inception of model-free and model-based learning via multi-voxel neurofeedback

Proposal/Award Number (if available):

Source of Support: NIH

Primary Place of Performance: University of Florida

Project/Proposal Support Start Date (if available): 07/2022

Project/Proposal Support End Date (if available): 06/2027

Total Award Amount (including Indirect Costs): \$3,169,468

Person-Month(s) (or Partial Person-Months) Per Year Committed to the Project:

Year	Person-months per year committed
------	----------------------------------

Year	Person-months per year committed
2023	1
2024	1
2025	1
2026	1
2027	1

Overall Objectives: To use neurofeedback to create associations between neural states and stimuli, linking stimuli to a common neural state in order to develop associations between them without participants having conscious knowledge of their pairing..

Statement of Potential Overlap: None with this proposal

Facilities, Equipment, and Other Resources

University of Florida (UF). UF is a public land-grant, sea-grant, and space-grant Research I (Carnegie Classification Very High Research) university, encompassing virtually all academic and professional disciplines. It is one of only 17 land-grant universities belonging to the Association of American Universities. The UF library system, including the Marston Science Library, serves as a valuable resource for the research and teaching missions of the investigators' respective departments, and a librarian is assigned to each of the investigators' respective departments to support faculty endeavors. UF has the capability of supporting the project described in this proposal.

Laboratory Space. Dr. Kvam has a suite of lab space including two sound-attenuated testing rooms (accommodating up to six participants at a time), a meeting room, and two offices for graduate researchers. This space holds six testing computers with touchscreen monitors (used to run touchscreen experiments but equally capable of serving as an ordinary monitor), an undergraduate workstation with touchscreen monitor, and a graduate workstation with modeling computer and touchscreen monitor. One of the testing computers is equipped with a Tobii Pro Nano eye tracker capable of recording pupil dilation and eye movements at a rate of 60 Hz with a precision of 0.10°. This space also includes locked filing cabinets for secure storage of consent forms and experiment-related materials.

High-performance computing Resources. Dr. Kvam has access to three specialized high-performance modeling computers as well as 12 supercomputing processor (NCU) cores through HiPerGator 3.0 (<https://www.rc.ufl.edu/services/hipergator/>) and an NVidia GPU, part of the A100 supercluster, for machine learning and modeling acceleration.

Conference / workshop resources. The University of Florida has a devoted Conference Department for planning the workshops proposed. This department assists in logistics, hotel contracts, travel reimbursements, A/V equipment and food & beverage menus. The University of Florida J. Wayne Reitz Union also provides room, equipment, and services to UF faculty for events (like the workshop we have proposed) at a discounted rate.

Other Resources. The Department of Psychology furnishes offices for Dr. Kvam and his graduate student and postdoctoral advisees, including desks, bookshelves, and telephones. The lab spaces have been outfitted with several locked filing cabinets for storing laptops, tablets, and any paper data while experiments are in progress. The Department has access to full-time professional research administrators to assist with grant administration in addition to two full-time IT professionals who help maintain the Department's computers. The Psychology Department maintains a participant pool to which we (as well as graduate researchers who will be funded by the grant) have access as members of the Department. Based on recent trends, we can each expect to be allotted up to 400 participants in Fall semesters and up to 300 in Spring semesters, more than sufficient for the needs of our unpaid / for-credit research studies.

Collaborators. Michael Gutter, the Associate Dean for Extension programs at the UF Institute of Food and Agricultural Science (IFAS), agreed to collaborate on the implementation of our risk and delay preference assessment system, its integration into Money Matters and First Time Homebuyers financial education program, and data collection on jobseekers' wage preferences through the IFAS Extension offices. Lisa Hamilton, Johanna Gomez-Gonzalez, Jenny Rodriguez, and Katherine Marin are all IFAS Extension Agents who carry out the financial education programs. They will collaborate on the project by recruiting participants, translating experimental materials and findings into Spanish, maintaining contact with participants to follow up after program participation, collaborate on dissemination of our findings (writing, presenting), and to integrate our preference assessment tools into ongoing financial education programs.

Data management plan

During the course of the proposed project, we will collect data from human subjects and perform secondary analyses on existing data. Experimental data will include price responses, self-report personality and behavior scales, response times, demographic data, and eye tracking data. In accordance with protections, health-related data will be anonymized and stored on a secure UF server so that they cannot be connected to participants.

All data we collect and analyze will be anonymized by coding each participant's data with a participant number. Each participant gets assigned a random number at the very beginning of each study, and they enter that number into the experiments and surveys in lieu of their name or any identifying information. This way, their name is never associated with their data and their name is only ever present on consent forms. For the repeated experiments where we follow up or bring participants in for multiple sessions, we keep a paper sheet with participants names and subject numbers in case participants forget their assigned number. This sheet will be stored in a locked filing cabinet in the laboratory space accessible only to study staff and destroyed upon conclusion of each experiment so that participant can no longer be connected to participants' names or identifying information. For human subjects data, consent will be obtained in accordance with the University of Florida Institutional Review Board, and all studies conducted will be subject to IRB oversight. All data collection will occur on the University of Florida campus in Gainesville or online through secure web servers hosted at UF. All data exchanged between these servers and researchers will be protected by Secured Sockets Layer encryption.

Physical or digital consent forms and debriefing will be provided to all participants, and signed consent forms will be stored in locked cabinets accessible only to study staff, and provided only upon request to entitled legal entities (such as the IRB or relevant government agencies). All data will be de-identified and stored electronically. The de-identified data will be stored as comma-separated value or MATLAB files and will be made publicly available upon submission of the research for consideration for publication on the Open Science Framework [OSF]. It will also be made available by request to other researchers. Model code will also be made publicly available on OSF and on University of Florida or PI's web servers. This code and data will be associated with readme / instruction files so that they are readily usable. Because the de-identified data will be stored on OSF, they will be retained indefinitely.

Each of the studies we run will be preregistered prior to data collection using either a Registered Report journal format or an open online registry like OSF Registries or AsPredicted. Prior to publication, research products related to this work will be provided as preprints on platforms such as PsyArXiv or arXiv.

We also expect that workshop materials will be developed during the course of the grant period, including but not limited to: model code, presentation slides, and other curriculum materials. These materials will be available to workshop participants and by request to the PI. They will also be used to improve the class materials in courses taught by the PI, graduate students, and postdocs sponsored by the grant.

Grant files and study documents will be retained for at least five fiscal years following the conclusion of the grant period in accordance with federal, state, and university regulations.



UF/IFAS Extension

Osceola County

1921 Kissimmee Valley Lane

Kissimmee, FL 34744

321-697-3000

321-697-3044 Fax

July 12th, 2022

To Whom It May Concern:

If the proposal submitted by Dr. Peter Kvam entitled "CAREER: An accessible, dynamic model of pricing applied to wage and housing purchases" is selected for funding by the NSF, it is my intent to collaborate and/or commit resources as detailed in the Project Description or the Facilities, Equipment or Other Resources section of the proposal.

Johanna Gomez Gonzalez

Johanna Gomez-Gonzalez

FCS Faculty Extension Agent I

UF/IFAS Extension Osceola County

The Foundation for The Gator Nation

An Equal Opportunity Institution



Institute of Food and Agricultural Sciences
UF/IFAS Extension Volusia County



3100 E. New York Avenue
DeLand, FL 32724
386-822-5778

July 13, 2022

Re: Letter of Collaboration

To Whom It May Concern:

If the proposal submitted by Dr. Peter Kvam entitled "CAREER: An accessible, dynamic model of pricing applied to wage and housing purchases" is selected for funding by the NSF, it is my intent to collaborate and/or commit resources as detailed in the Project Description or the Facilities, Equipment or Other Resources section of the proposal.

With regards,

A handwritten signature in dark ink that reads "Lisa A. Hamilton". The signature is written in a cursive, flowing style.

Lisa Hamilton
Family and Consumer Science Extension Agent
UF/IFAS Extension Volusia County



Institute of Food and Agricultural Science
Orange Cooperative Extension Service



6021 S. Conway Rd.
Orlando, FL 32812
Tel: (407) 254 9231

July 14, 2022

Ref: Letter of Collaboration

To Whom It May Concern:

If the proposal submitted by Dr. Peter Kvam entitled "CAREER: An accessible, dynamic model of pricing applied to wage and housing purchases" is selected for funding by the NSF, it is my intent to collaborate and/or commit resources as detailed in the Project Description or the Facilities, Equipment or Other Resources section of the proposal.

With kind regards,

Jenny Aviles Rodriguez

*Jenny Aviles Rodriguez, Agent I –
Family & Consumer Sciences*
University of Florida IFAS Extension Orange County
6021 S. Conway Rd. Orlando, FL 32812
Tel: (407) 254 9231
Fax: (407) 850 5125
jennyarodriguez@ufl.edu

John D. Campbell Agricultural Center
18710 SW 288th Street
Homestead, FL 33030
Telephone: (305) 248-3311

July 14, 2022

Ref: Letter of Collaboration

To Whom It May Concern:

If the proposal submitted by Dr. Peter Kvam entitled "CAREER: An accessible, dynamic model of pricing applied to wage and housing purchases" is selected for funding by the NSF, it is my intent to collaborate and/or commit resources as detailed in the Project Description or the Facilities, Equipment or Other Resources section of the proposal.

With kind regards,



Katherine Marin, M.Ed., MHFA
Community Development, Extension Agent
University of Florida IFAS Extension
18710 SW 288 St., Homestead, FL 33030
C: 305-679-0223 | O: 305-248-3311



College of Liberal Arts & Sciences
Department of Psychology

945 Center Dr
PO Box 112250
Gainesville, FL 32611-0000
352-273-2139

July 12, 2022

To whom it may concern:

This letter expresses my enthusiastic support for Dr. Peter Kvam's application to the National Science Foundation's CAREER Program in the Decision, Risk, and Management Sciences area and the research program it entails. As a tenure-track member of our faculty, the University of Florida, College of Liberal Arts & Science, and the Department of Psychology have made a long-term commitment to the support and development of his research, teaching, and advising. His track record of research before and since arriving at the University has been excellent, which is reflected in his annual reviews and feedback from merit and mentoring committees. In particular, he is seen as a rising star in computational modeling and decision making research, and has been at the forefront of the department's contributions to the University of Florida's Artificial Intelligence initiative.

As a recently hired tenure-track faculty member, Dr. Kvam has been provided with 9-month salary support for the duration of the project, with teaching releases (a 1-1 load for his first two years at UF; 2-1 in subsequent years, including the proposed grant period), dedicated time for research (currently 71% research effort), and funding for graduate students who work in his laboratory. Dr. Kvam is also provided guidance through a formal mentoring committee consisting of four senior faculty in the College of Liberal Arts & Sciences as well as the College of Health & Human Performance, all of whom have had previous experience as Principal Investigators on similar research grants. The University of Florida also provides free training on research ethics, institutional review board, and grant management through regular workshops and through the College of Liberal Arts & Sciences Research Office. The Department of Psychology also provides access to administrative staff for grant, travel, and budget management as well as information technology staff for equipment acquisition, training, and technical support.

Dr. Kvam has been provided with laboratory and office space as well as start-up funds to support establishing the lab and conducting pilot studies. This includes a suite of lab space including two sound-attenuated testing rooms, an adjoining meeting room, two offices for laboratory staff, a participant reception area, and an office for himself equipped with phone, bookshelves, desk, and computer. His start-up funding has been used to purchase laboratory furniture and equipment, support travel to professional conferences, cover open-access publishing fees, support membership in professional societies, and to purchase experimental testing computers, high-performance modeling computers for model fitting, and a research computing investment in HiPerGator, the University of Florida's supercomputing cluster. HiPerGator 3.0 and HiPerGator AI, including the NVIDIA DGX A100 SuperPod, are supported by the UF Research Computing (UFRC) team, who maintain this cluster and support training in research on Artificial Intelligence. Training on deep

learning and AI through UFRC and NVIDIA will be available to Dr. Kvam as well as to the postdoctoral and graduate student researchers on the project.

Notably, in addition to start-up funds to establish his lab and fund initial studies, Dr. Kvam has also received competitive institutional funding from the UF Informatics Institute including a SEED grant as PI and as the mentor of a graduate fellow with the institute.

In addition to research and training of graduate students, Dr. Kvam is also making important contributions to the department's educational mission. He has developed a new course in our undergraduate curriculum, Decisions & Judgment, that introduces students to multi-disciplinary perspectives on decision making; he has also re-designed an upper-level undergraduate lab course that incorporates training in R, MATLAB, and Python and gives students hands-on opportunities to apply programming skills to studies of sensory processing. At the graduate level, he has developed a new course on Psychological & Behavioral computational modeling and offered a workshop on Bayesian statistics. Finally, his lab includes a large number of undergraduate students who contribute to the team science of the lab with many students included on conference presentations and publications. Hence, there is substantial synergy in Dr. Kvam's research and teaching contributions which has expanded the opportunities for our students to gain experience in innovative research and also gain transferrable skills in both the classroom and lab.

Given Dr. Kvam's research productivity to date and innovative research ideas, I am confident that he will continue on his current trajectory making high impact contributions to computational modeling of decision making processes. Moreover, I am confident that he will establish an externally funded program of research typical of the top scientists in the field.

Sincerely,



Julia A. Graber, Ph.D.
Professor & Chair, Department of Psychology

Table 1

1	Your Name:	Your Organizational Affiliation(s), last 12 mo	Last Active Date
	Kvam, Peter D	University of Florida	

Table 2

2	Name:	Type of Relationship	Optional (email, Department)	Last Active Date
R	Kvam, Paul H	Family	Department of Statistics, University of Richmond	
R	Kvam, Eric P	Family	Dept of Materials Engineering, Purdue University	

Table 3

3	Advisor/Advisee Name:	Organizational Affiliation	Optional (email, Department)
G	Pleskac, Timothy J	University of Kansas	Department of Psychology
T	Sokratous, Konstantina	University of Florida	Department of Psychology
T	Alaukik, Abhay	University of Florida	Department of Psychology
T	Anderson, Fitch	University of Florida	Department of Psychology

Table 4

4	Name:	Organizational Affiliation	Optional (email, Department)	Last Active Date
A	Turner, Brandon M	The Ohio State University, UNITED STATES		
A	Busemeyer, Jerome R	Indiana University, UNITED STATES		
A	Pleskac, Timothy J	University of Kansas, UNITED STATES		
A	Hertwig, Ralph	Max Planck Institute for Human Development, GERMANY		
A	Pachur, Thorsten	Max Planck Institute for Human Development, GERMANY		
A	Hintze, Arend	Dalarna University, SWEDEN		
A	Vassileva, Jasmin	Virginia Commonwealth University, UNITED STATES		
A	Finn, Peter	Indiana University, UNITED STATES		
A	Romeu, Ricardo	Indiana University, UNITED STATES		
A	Heathcote, Andrew	University of Tasmania, AUSTRALIA		
A	Reynolds, Angus	University of Tasmania, AUSTRALIA		
A	Sauer, James	University of Tasmania, AUSTRALIA		
A	Garton, Roderick	University of Tasmania, AUSTRALIA		

A	Molloy, M Fiona	The Ohio State University, UNITED STATES		
A	Osth, Adam	University of Melbourne, AUSTRALIA		
A	Haines, Nathaniel	The Ohio State University, UNITED STATES		
A	Blanco, Nathaniel J	The Ohio State University, UNITED STATES		
A	Unger, Layla	The Ohio State University, UNITED STATES		
A	Ralston, Robert	The Ohio State University, UNITED STATES		
A	Beauchaine, Theodore	The Ohio State University, UNITED STATES		
A	Pitt, Mark	The Ohio State University, UNITED STATES		
A	Sloutsky, Vladimir	The Ohio State University, UNITED STATES		
A	Griffin, Valerie	University of Tasmania, AUSTRALIA		
A	Rahnev, Dobromir	Georgia Institute of Technology, UNITED STATES		
A	Ahn, Woo Young	Seoul National University, SOUTH KOREA		
A	Hawkins, Guy	University of Newcastle, AUSTRALIA		
A	Baldwin, Matthew	University of Florida, UNITED STATES		
A	Mims, Callie	University of Southern Alabama, UNITED STATES		
A	Irving, Louis	University of Florida, UNITED STATES		
A	Smith, Colin	University of Florida, UNITED STATES		
A	Westgate, Erin	University of Florida, UNITED STATES		
C	Karmakar, Sayar	University of Florida, UNITED STATES		
C	Duan, Leo Li	University of Florida, UNITED STATES		

Table 5

5	Name:	Organizational Affiliation	Journal/Collection	Last Active Date
B	Rotello, Caren	University of Massachusetts Amherst	Cognitive Psychology	

List of Suggested Reviewers

Data Not Available

List of Reviewers Not to Include

Data Not Available